

Edge-Subset Lattice and Its Application to Linear Network Error Correction Coding

Xuan Guang^{ID}, Member, IEEE, Ruopu Cui^{ID}, and Raymond W. Yeung^{ID}, Fellow, IEEE

Abstract—In this paper, we first explore the underlying mathematical structure of edge subsets on a finite directed acyclic graph in using a lattice-theoretic approach. We prove that a collection of edge subsets, depending on different conditions it satisfies, together with the corresponding “cut-separating” partial order, can form a meet-semilattice, a join-semilattice or a lattice. The bottom and top thus derived generalize the concept of the primary minimum cut introduced by Guang and Yeung (2018) and hence we provide a new way from the lattice-theoretic point of view to understand the primary minimum cut and its existence and uniqueness. The introduced concepts and obtained results regarded as a bridge connect graph theory and lattice theory, which appear to be of fundamental interest in graph theory, lattice theory, and even beyond. In addition, we consider linear network error correction (LNEC) coding when errors may occur on edges of a communication network of which the topology is known. In LNEC coding, the minimum required field size for the existence of LNEC codes, in particular LNEC maximum distance separable (MDS) codes which are regarded as a most important type of optimal codes, is an open problem not only of theoretical interest but also of practical importance. By applying the approach of the above edge-subset (semi-)lattice, we obtain an improved upper bound on the minimum required field size for the existence of LNEC (MDS) codes. We quantify the improvement over the existing results by both theoretical analysis and numerical simulations and thus show that the improvement is in general significant. The improved upper bound, which is graph-theoretic, depends only on the network topology and requirement of the error correction capability but not on a specific code construction. We also develop an efficient algorithm that can compute the bound in a linear time of the number of edges.

Index Terms—Edge-subset (semi-)lattice, linear network error correction (LNEC) coding, primary minimum cut, top/bottom,

the minimum required field size for the existence of LNEC codes, graph-theoretic algorithms, lattice theory, graph theory.

I. INTRODUCTION

IN 1956, the problem of maximizing the rate of flow from a source node to a sink node through a network was considered independently by Elias et al. [1] and Ford and Fulkerson [2], where, regardless of whether the flow is a commodity flow or an information flow, the value of the maximum flow is equal to the capacity of a minimum cut separating the sink node from the source node. This result is the celebrated *max-flow min-cut theorem*, proved in [1] and [2]. In 2000, Ahlswede et al. [3] put forward the general concept of *network coding* that allows the intermediate nodes in a noiseless network to process the received information. In particular, they focused on the single-source network coding problem on a general network and proved that if coding (rather than routing only) is applied at the nodes in a network, the single source node can multicast messages to all the sink nodes at the theoretically maximum rate, i.e., the smallest minimum cut capacity between the source node and a sink node, as the alphabet size of both the information source and the channel transmission symbol tends to infinity. This result can be regarded as the max-flow min-cut theorem for information flow from a source node multicasting to multiple sink nodes through a network, as well as a generalization of the classical max-flow min-cut theorem from a source node to a sink node through a network.

A. Motivation of the Paper

In the paradigm of network coding, network error correction is necessary when errors may occur on the links of a communication network. In general, the problem induced by errors in network coding can be more serious than the one in a classical point-to-point communication system, because errors will be propagated by the coding operations at the intermediate nodes. Even a single error occurred on an edge has the potential of polluting all the “downstream” messages. The network coding techniques for combating network errors are referred to as the *network error correction coding*. Further, linear network coding techniques for combating network errors are referred to as *linear network error correction (LNEC) coding*, which was investigated widely in the literature, e.g., [4], [5], [6], [7], [8], [9], [10], [11], [12], [13], [14], [15], [16], [17], [18]. In particular, we consider a very special case of network error correction coding over the simplest network as depicted in

Received 23 May 2024; revised 13 December 2024; accepted 22 March 2025. Date of publication 31 March 2025; date of current version 18 June 2025. This work was supported in part by the National Key Research and Development Program of China under Grant 2023YFA1009604 and Grant 2022YFA1005000; in part by the Natural Science Foundation of China under Grant 62171238 and Grant 62461160306; in part by the Fundamental Research Funds for Central Universities of China under Grant NKU 050-63253087; in part by the Research Grants Council of Hong Kong, SAR, China, under Grant 14207219; and in part by the Fellowship Award from the Research Grants Council of Hong Kong Special Administrative Region, China, under Grant CUHK SRFS2223-4S03. (Corresponding author: Xuan Guang.)

Xuan Guang and Ruopu Cui are with the School of Mathematical Sciences and the Key Laboratory of Pure Mathematics and Combinatorics (LPMC), Nankai University, Tianjin 300071, China (e-mail: xguang@nankai.edu.cn; rpcui@mail.nankai.edu.cn).

Raymond W. Yeung is with the Institute of Network Coding and the Department of Information Engineering, The Chinese University of Hong Kong, Hong Kong, SAR, China (e-mail: whyeung@ie.cuhk.edu.hk).

Communicated by F. Manganiello, Associate Editor for Shannon Theory and Information Measures.

Color versions of one or more figures in this article are available at <https://doi.org/10.1109/TIT.2025.3556551>.

Digital Object Identifier 10.1109/TIT.2025.3556551

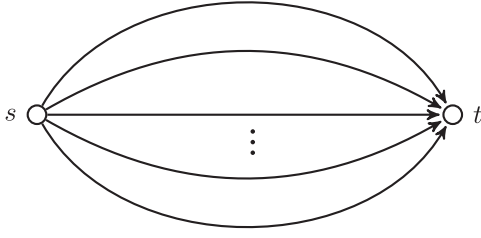


Fig. 1. An equivalent model of the classical coding theory.

Fig. 1, which consists of only two nodes, a source node s and a sink node t , connected by multiple edges from s to t . This special case of network error correction coding can be regarded as the model of classical coding theory (cf. [19], [20]), a very rich field of research originated from Shannon's seminal work [21] in 1948.

Network error correction coding was first considered by Cai and Yeung [4], and they further developed network error correction coding in their two-part paper [5], [6] as a generalization of algebraic coding from the point-to-point setting to the network setting. Subsequently, two approaches of LNEC coding were developed in [7] and [8]. Recently, Guang and Yeung [9] present the framework of additive adversarial network for LNEC coding. By using this framework with the obtained results, the above two LNEC approaches, studied in [7], [10] and [8], [11], respectively, are unified under this framework and proved to be equivalent. The construction of LNEC codes was also investigated in the literature, such as [10], [11], [12], [13], and [14].

In LNEC coding, the minimum required field size for the existence of LNEC codes, in particular LNEC maximum distance separable (MDS) codes which are regarded as a most important type of optimal codes that achieve the Singleton bound with equality, is an open problem not only of theoretical interest but also of practical importance, because it is closely related to the implementation of the coding scheme in terms of computational complexity and storage requirement [5], [6], [7], [9], [10], [11], [12], [13], [14], [15], [16]. However, the existing upper bounds on the minimum required field size for the existence of LNEC (MDS) codes, are typically too large for the implementation. In this paper, we explore the underlying mathematical structure of edge subsets on a directed acyclic graph in a lattice-theoretic approach. By using the obtained (semi-)lattice structure of edge subsets and the developed approach, we show that the required field size for the existence of LNEC (MDS) codes can be reduced significantly.

B. Contributions and Organization of the Paper

The main contributions and organization of the paper are given as follows:

- In Section II, we present the network model and introduce the necessary notation and definitions.
- In Section III, we explore the underlying mathematical structure of edge subsets on a finite directed acyclic graph using a lattice-theoretic approach. We prove that a collection of edge subsets, depending on different conditions it satisfies, together with the corresponding “cut-separating”

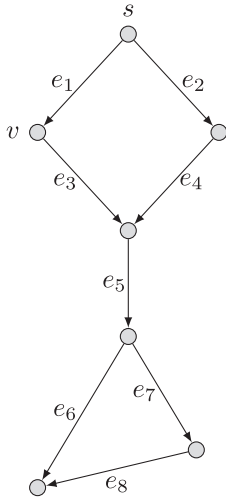
partial order, can form a meet-semilattice, a join-semilattice or a lattice. The bottom and the top thus induced generalize the concept of the primary minimum cut introduced by Guang and Yeung in [22, Section IV-A], and hence we provide a new way from the lattice-theoretic point of view to understand the primary minimum cut and its existence and uniqueness. Further, by modifying the algorithm for finding the primary minimum cut (Algorithm 2 in [22]), we develop efficient algorithms for finding the top and/or the bottom of an edge-subset (semi-)lattice.

- In Section IV, by applying the obtained results on edge-subset (semi-)lattices, we prove an improved upper bound on the minimum required field size, which is the total number of tops of the join-semilattices for every sink node. We quantify the improvement over the existing results by both theoretical analysis and numerical simulations, which show that the improvement is in general significant. This improved bound, which is graph-theoretic, depends on the network topology and the requirement of the error correction capability but not on the specific code construction. With the algorithms for finding the top and/or the bottom of the (semi-)lattice, we then develop an efficient algorithm for computing this bound.
- We conclude in Section V with a summary of our results and a remark on future research.
- The notations frequently used in the paper are listed in Table I (see Appendix A).

II. NETWORK MODEL

We use a finite acyclic graph to model a communication network. Let $G = (V, E)$ be a finite directed acyclic graph with a single source s and a set of sink nodes $T \subseteq V \setminus \{s\}$, where V and E are the sets of nodes and edges, respectively. For a directed edge e from node u to node v , the *tail* u and the *head* v of an edge $e \in E$ are denoted by $\text{tail}(e)$ and $\text{head}(e)$, respectively. For example, consider the edge e_1 in the graph G depicted in Fig. 2 below. The tail and the head of e_1 are s and v , respectively, i.e., $\text{tail}(e_1) = s$ and $\text{head}(e_1) = v$. Further, for a node v , let $\text{Out}(v) = \{e \in E : \text{tail}(e) = v\}$ and $\text{In}(v) = \{e \in E : \text{head}(e) = v\}$, which are the set of output edges and the set of input edges, respectively. Without loss of generality, assume that there are no input edges for the source node s and no output edges for any sink node $t \in T$. The capacity of each edge is taken to be 1, i.e., a symbol taken from an alphabet is transmitted on each edge $e \in E$ for each use of e . Further, parallel edges between two adjacent nodes are allowed.

In the graph G , if a sequence of edges $(e_1, e_2, \dots, e_m)$ satisfies $\text{tail}(e_{k+1}) = \text{head}(e_k)$ for all $k = 1, 2, \dots, m-1$, then $(e_1, e_2, \dots, e_m)$ is called a *path* from the node $\text{tail}(e_1)$ (or the edge e_1) to the node $\text{head}(e_m)$ (or the edge e_m). In particular, a single edge e is regarded as a path from $\text{tail}(e)$ to $\text{head}(e)$ (or from e to itself). For two nodes u and v , a *cut* separating v from u is a set of edges whose removal disconnects v from u , i.e., no path exists from u to v upon deleting the edges in this set. The *capacity* of this cut separating v from u is defined as the number of edges in the cut. The minimum of the

Fig. 2. The graph G .

capacities of all cuts separating v from u is called the *minimum cut capacity* separating v from u . Further, a cut is called a *minimum cut* separating v from u if its capacity achieves this minimum cut capacity. If u and v are two nodes such that v is disconnected from u , i.e., no path exists from u to v in the graph G , we adopt the convention that the minimum cut capacity separating v from u is 0 and the empty set of edges is the minimum cut separating v from u .

These concepts can be extended from separating a node v from another node u to separating a nonempty subset of nodes $\widehat{V}$ from a node u ($u \notin \widehat{V}$), and separating an edge subset ξ from a node u as follows. We first consider a nonempty subset of non-source nodes $\widehat{V} \subseteq V$. We create a new node $v_{\widehat{V}}$, and for every node v in $\widehat{V}$, add a “super-edge” of infinite capacity from v to $v_{\widehat{V}}$ (which is equivalent to adding an infinite number of parallel edges from v to $v_{\widehat{V}}$). A *cut* separating $\widehat{V}$ from u is defined as a cut of *finite* capacity separating $v_{\widehat{V}}$ from u .¹ We can naturally extend the definitions of the capacity of a cut, the minimum cut capacity, and the minimum cut to the case of $\widehat{V}$. We further consider an edge subset $\xi \subseteq E$. We first subdivide each edge $e \in \xi$ by creating a node v_e and splitting e into two edges $e^{(1)}$ and $e^{(2)}$ such that $\text{tail}(e^{(1)}) = \text{tail}(e)$, $\text{head}(e^{(2)}) = \text{head}(e)$, and $\text{head}(e^{(1)}) = \text{tail}(e^{(2)}) = v_e$. Let $V_\xi = \{v_e : e \in \xi\}$. Then a *cut* separating the edge subset ξ from the node u is defined as a cut separating V_ξ from u , where, whenever $e^{(1)}$ or $e^{(2)}$ appears in the cut, replace it by e . In particular, by definition, ξ is a cut separating ξ itself from u . To be specific, we readily see that $\{e^{(1)} : e \in \xi\}$ is a cut separating V_ξ from u , and then by definition, we replace each $e^{(1)}$ in $\{e^{(1)} : e \in \xi\}$ by the corresponding edge e to obtain the edge subset ξ , which thus is a cut separating ξ from u . Similarly, the minimum cut capacity separating V_ξ from u is defined as the *minimum cut capacity* separating ξ from u . Also, a cut separating ξ from u achieving this minimum cut capacity is called a *minimum cut* separating ξ from u . If an edge set $A \subseteq E$ is a cut separating a node v (resp. a set of nodes $\widehat{V}$ and a set

¹An equivalent definition of a cut separating $\widehat{V}$ from u is a set of edges satisfying that no path exists from u to every v in $\widehat{V}$ upon deleting the edges in this set. The case of the former definition facilitates the following discussion of the paper.

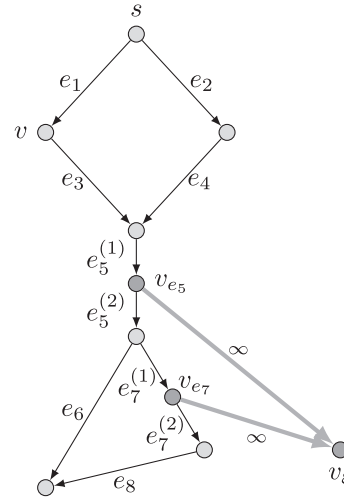


Fig. 3. The graph modification.

of edges ξ) from another node u , then we say that the edge set A *separates* v (resp. $\widehat{V}$ and ξ) *from* u . Now, we use a graph in [23] (Figs. 2 and 3) as an example to illustrate the above graph-theoretic concepts for the case of an edge subset.

Example 1: We consider an edge subset $\xi = \{e_5, e_7\}$ in the graph G depicted in Fig 2, where s is the source node and the capacity of edge is taken to be 1. By an edge subdivision operation, the edge e_5 is split into two edges $e_5^{(1)}$ and $e_5^{(2)}$ in Fig. 3, with $\text{tail}(e_5^{(1)}) = \text{tail}(e_5)$, $\text{head}(e_5^{(2)}) = \text{head}(e_5)$, and $\text{head}(e_5^{(1)}) = \text{tail}(e_5^{(2)}) = v_{e_5}$. In the same way, the edge e_7 is split into two edges $e_7^{(1)}$ and $e_7^{(2)}$. Let $V_\xi = \{v_{e_5}, v_{e_7}\}$. Now, in order to find a cut separating ξ from s , it is equivalent to finding a cut separating V_ξ from s . Toward this end, we first create a new node v_ξ and add two super-edges of infinite capacity from v_{e_5} to v_ξ and from v_{e_7} to v_ξ , respectively (cf. Fig 3). By definition, a cut of finite capacity separating v_ξ from s is a cut separating V_ξ from s and so a cut separating ξ from s . For example, the edge subset $\{e_3, e_4\}$ is such a cut. Further, we consider the edge subset $\{e_5^{(1)}\}$, which is also a cut separating v_ξ from s . By definition, since $e_5^{(1)}$ appears in the cut $\{e_5^{(1)}\}$ and $e_5 \in \xi$, we replace $e_5^{(1)}$ by e_5 and say that $\{e_5\}$ is a cut separating ξ from s . In particular, $\{e_5\}$ is also a minimum cut separating ξ from s and thus the minimum cut capacity separating ξ from s is 1.

We continue to consider a finite directed acyclic graph $G = (V, E)$. We define the *reverse graph* $G^\top$ of G that is obtained from G by reversing the direction of every edge on G . It is evident that an edge subset of E is a cut separating a node v from a node u on G if and only if this edge subset of E with reversed direction is a cut separating u from v on $G^\top$. Inspired by this observation, for a nonempty subset of nodes $\widehat{V}$ (resp. a nonempty edge subset ρ) and a node u , we say that an edge subset of E is a *cut separating* u *from* $\widehat{V}$ (resp. ρ) on G if this edge subset (with reversed direction) is a cut separating $\widehat{V}$ (resp. ρ) from u on $G^\top$. We can naturally extend the definitions of the capacity of a cut, the minimum cut capacity, and the minimum cut to the cases of $\widehat{V}$ and ρ . In particular, we elaborate these definitions for the case of edge subset ρ , which will be frequently used in the following

discussions of the paper. The *capacity* of a cut separating u from ρ on G is defined as the number of edges in the cut. The minimum of the capacities of all cuts separating u from ρ on G is called the *minimum cut capacity* separating u from ρ , denoted by $\text{mincut}(\rho, u)$. On G , a cut separating u from ρ is called a *minimum cut* separating u from ρ if its capacity achieves the minimum cut capacity $\text{mincut}(\rho, u)$.

III. SEMILATTICE AND LATTICE OF EDGE SUBSETS

A. Semilattice and Lattice

In this subsection, we present some concepts and properties on lattice in order theory (cf. [24], [25]).

Definition 1: Let P be a nonempty set. A (non-strict) partial order “ $\leq$ ” on P is a binary relation among the elements in P that satisfies the following conditions for arbitrary elements a, b and c in P ,

- 1) **(Reflexivity)** $a \leq a$;
- 2) **(Antisymmetry)** if $a \leq b$ and $b \leq a$, then $a = b$;
- 3) **(Transitivity)** if $a \leq b$ and $b \leq c$, then $a \leq c$.

The set P associated with the partial order “ $\leq$ ” is called a *partially ordered set (poset)*, which is written as $(P, \leq)$, or P for simplicity when the partial order “ $\leq$ ” is clear from the context. For a partially ordered set $(P, \leq)$, we consider a subset $Q \subseteq P$. An element $a \in Q$ is a *maximal* element of Q if $a \leq x$ and $x \in Q$ imply $a = x$. Likewise, an element $b \in Q$ is a *minimal* element of Q if $x \leq b$ and $x \in Q$ imply $b = x$. Furthermore, an element $a \in P$ is an *upper bound* of Q if $x \leq a$ for all $x \in Q$. A *lower bound* of Q is similarly defined. The set of all upper bounds of Q and the set of all lower bounds of Q are denoted by Q^u and Q^l , respectively, i.e.,

$$Q^u = \{a \in P : x \leq a, \forall x \in Q\}$$

and

$$Q^l = \{a \in P : a \leq x, \forall x \in Q\}.$$

An upper bound a^* of Q is called the *supremum (least upper bound)* if $a^* \leq a$ for all $a \in Q^u$. Similarly, we can define the *infimum (greatest lower bound)* of Q . We remark that the supremum (infimum) may not exist.

From the algebraic point of view, the supremum of a subset $Q \subseteq P$ is called the *join* of Q , denoted by $\vee Q$; and the infimum of Q is called the *meet* of Q , denoted by $\wedge Q$. In particular, for a two-element subset $\{a, b\} \subseteq P$, we write the join $\vee\{a, b\}$ as $a \vee b$ and the meet $\wedge\{a, b\}$ as $a \wedge b$. Accordingly, the join “ $\vee$ ” and the meet “ $\wedge$ ” can be regarded as two binary operations on two elements in P .

Definition 2: A partially ordered set $(P, \leq)$ is a *join-semilattice* (or *upper semilattice*) if the join exists for any two elements in P . Similarly, $(P, \leq)$ is a *meet-semilattice* (or *lower semilattice*) if the meet exists for any two elements in P . Further, a partially ordered set $(P, \leq)$ is a *lattice* if it is both a join-semilattice and a meet-semilattice, i.e., both the join and the meet exist for any two elements in P .

By the above definition, we immediately have the following properties for semilattices and lattice, which can be easily proved by induction.

Proposition 1: For a join-semilattice (meet-semilattice) P , the join $\vee Q$ (the meet $\wedge Q$) exists for any finite subset Q of P .

For a lattice P , both the join $\vee Q$ and the meet $\wedge Q$ exist for any finite subset Q of P .

Definition 3: Let $(P, \leq)$ be a partially ordered set. An element in P is called a *bottom*, denoted by $\perp$, if $\perp \leq a$ for all $a \in P$; and dually, an element in P is called a *top*, denoted by $\top$, if $a \leq \top$ for all $a \in P$.

By definition, if a partially ordered set P has a bottom (or a top), the bottom (or the top) is unique. Immediately, we have the following proposition.

Proposition 2: A finite join-semilattice has a (unique) top $\top$, and dually, a finite meet-semilattice has a (unique) bottom $\perp$. Then a finite lattice has both a (unique) top $\top$ and a (unique) bottom $\perp$.

B. Semilattice and Lattice of Edge Subsets

Let $G = (V, E)$ be a directed acyclic graph with unit capacity for each edge, where parallel edges between two adjacent nodes are allowed. Let $s \in V$ and $t \in V$ be a source node without input edges and a sink node without output edges, respectively. We consider a natural ancestral topological order “ $\leq$ ” on the edges in E as follows: for two edges d and e in E , we write $d \leq e$ if either $d = e$ or there exists a directed path from d to e . We readily see that the binary relation “ $\leq$ ” is a non-strict partial order on E .

Next, we consider an edge subset ρ on G , and denote by $\text{mincut}(s, \rho)$ the minimum cut capacity separating ρ from s . If $|\rho| = \text{mincut}(s, \rho)$, then we say that the edge subset ρ is *regular from the source node s* , or *s-regular* for simplicity. For two edge subsets ρ and η , we say that ρ and η has a binary relation “ $\sim_s$ ”, written as $\rho \sim_s \eta$, if and only if there exists an edge subset ξ that is a minimum cut separating ρ from s and also separating η from s , namely that, ρ and η have a common minimum cut ξ separating each of them from s . Further, if ρ and η are s -regular, then $\rho \sim_s \eta$ implies $|\rho| = |\eta|$ because $|\rho| = \text{mincut}(s, \rho) = |\xi| = \text{mincut}(s, \eta) = |\eta|$. The binary relation “ $\sim_s$ ” was introduced by Guang and Yeung in [22], where it was proved that “ $\sim_s$ ” is an equivalence relation.

Next, we consider the collection $\mathcal{B}_s(r)$ of all the s -regular edge subsets of size r , i.e.,

$$\mathcal{B}_s(r) = \{\rho \subseteq E : |\rho| = \text{mincut}(s, \rho) = r\}, \quad (1)$$

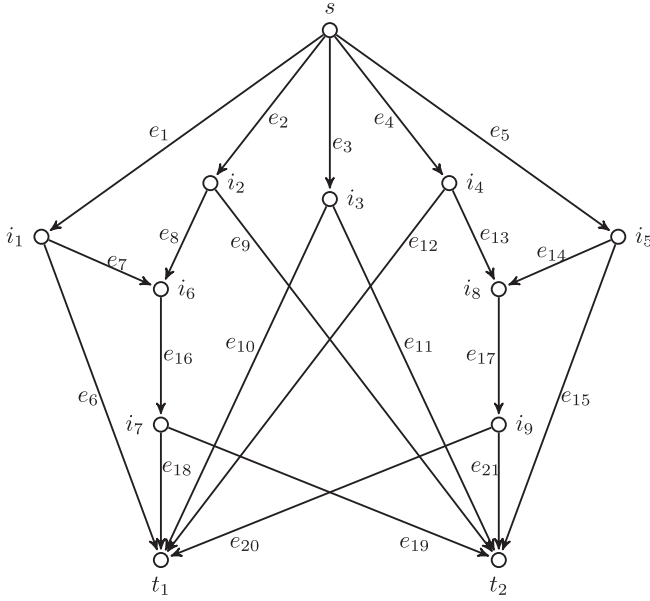
where r is an arbitrary positive integer such that $\mathcal{B}_s(r) \neq \emptyset$.

We now define another binary relation “ $\leq_s$ ” among the edge subsets in $\mathcal{B}_s(r)$ as follows: for two s -regular edge subsets ρ and η in $\mathcal{B}_s(r)$, we write $\rho \leq_s \eta$ if no path exists from s to any edge in η upon deleting the edges in ρ . Before discussing further, we give an example to illustrate the above graph-theoretic concepts.

Example 2: Consider the graph $G = (V, E)$ as depicted in Fig. 4, and two edge subsets $\rho \triangleq \{e_1, e_8\}$ and $\eta \triangleq \{e_6, e_{16}\}$. We first note that ρ and η are s -regular because

$$|\rho| = \text{mincut}(s, \rho) (= 2) \quad \text{and} \quad |\eta| = \text{mincut}(s, \eta) (= 2).$$

Next, consider the edge subset $\xi \triangleq \{e_1, e_2\}$. We see that ξ is a common minimum cut separating ρ and η from the source node s , and thus $\rho \sim_s \eta$. Further, upon deleting the edges e_1 and e_8 (the two edges in ρ) from G , no path exists from s to e_6 or

Fig. 4. The graph G .

e_{16} (the two edges in η). Thus, by definition, we immediately obtain that $\rho \leq \eta$.

The next lemma asserts that the binary relation “ $\leq$ ” is also a partial order.

Lemma 1: The binary relation “ $\leq$ ” on $\mathcal{B}_s(r)$ is a partial order.

Proof: Reflexivity and transitivity are apparent. For anti-symmetry, we assume that $\rho \leq \eta$ and $\eta \leq \rho$ for two s -regular edge subsets ρ and η in $\mathcal{B}_s(r)$. By the definition of the binary relation “ $\leq$ ”, we obtain that ρ separates η from s and η separates ρ from s . This implies that $\rho = \eta$. The lemma is proved. ■

With the equivalence relation “ $\sim$ ”, the collection $\mathcal{B}_s(r)$ is partitioned into distinct equivalence classes. We now present the following theorem, a main result in this section, which asserts that an equivalence class of $\mathcal{B}_s(r)$ partially ordered by the binary relation “ $\leq$ ” forms a meet-semilattice.

Theorem 2: Let Cl_r be an equivalence class of $\mathcal{B}_s(r)$. Then $(\text{Cl}_r, \leq)$ is a meet-semilattice.

To prove Theorem 2, we need several concepts and some results.

Lemma 3 ([22, Proposition 13]): Let $n = \text{mincut}(s, t)$ and CUT be an arbitrary minimum cut between s and t . Then an arbitrary set of n edge-disjoint paths from s to t contains all the n edges in CUT , and each of the n edges is on exactly one of the n edge-disjoint paths.

Let $n = \text{mincut}(s, t)$ and $\mathcal{P} = \{P_1, P_2, \dots, P_n\}$ be a set of n edge-disjoint paths $P_1, P_2, \dots, P_n$ from s to t . Consider two minimum cuts CUT_1 and CUT_2 separating t from s . By Lemma 3, we let $P_i \cap \text{CUT}_1 = \{e_{1,i}\}$ and $P_i \cap \text{CUT}_2 = \{e_{2,i}\}$ for $1 \leq i \leq n$, i.e.,

$$\text{CUT}_1 = \{e_{1,i} : i = 1, 2, \dots, n\}$$

and

$$\text{CUT}_2 = \{e_{2,i} : i = 1, 2, \dots, n\}.$$

For each $i = 1, 2, \dots, n$, we define

$$\text{minord}(e_{1,i}, e_{2,i}) = \begin{cases} e_{1,i}, & \text{if } e_{1,i} \leq e_{2,i}, \\ e_{2,i}, & \text{otherwise;} \end{cases} \quad (2)$$

and

$$\text{maxord}(e_{1,i}, e_{2,i}) = \begin{cases} e_{2,i}, & \text{if } e_{1,i} \leq e_{2,i}, \\ e_{1,i}, & \text{otherwise;} \end{cases}$$

and accordingly we further define two edge subsets

$$\text{CUT}_{\min}^{\mathcal{P}} \triangleq \{\text{minord}(e_{1,i}, e_{2,i}) : i = 1, 2, \dots, n\} \quad (3)$$

and

$$\text{CUT}_{\max}^{\mathcal{P}} \triangleq \{\text{maxord}(e_{1,i}, e_{2,i}) : i = 1, 2, \dots, n\}. \quad (4)$$

The following theorem shows that the two edge subsets $\text{CUT}_{\min}^{\mathcal{P}}$ and $\text{CUT}_{\max}^{\mathcal{P}}$ are still minimum cuts separating t from s , and independent of which set of n edge-disjoint paths from s to t is chosen.

Theorem 4: Let $n = \text{mincut}(s, t)$. Consider two minimum cuts CUT_1 and CUT_2 separating t from s , and a set $\mathcal{P}$ of n edge-disjoint paths $P_1, P_2, \dots, P_n$ from s to t . Then, the edge subsets $\text{CUT}_{\min}^{\mathcal{P}}$ and $\text{CUT}_{\max}^{\mathcal{P}}$ are also two minimum cuts separating t from s . Furthermore, for any other set $\mathcal{P}'$ of n edge-disjoint paths from s to t ,

$$\text{CUT}_{\min}^{\mathcal{P}'} = \text{CUT}_{\min}^{\mathcal{P}} \quad \text{and} \quad \text{CUT}_{\max}^{\mathcal{P}'} = \text{CUT}_{\max}^{\mathcal{P}}. \quad (5)$$

We need the following two lemmas to prove Theorem 4.

Lemma 5 ([22, Lemma 14]): Let $\text{MinCut}(s, t)$ be the set of all the minimum cuts separating t from s . For any two minimum cuts $\text{CUT}_1, \text{CUT}_2 \in \text{MinCut}(s, t)$, $\text{CUT}_1 \leq \text{CUT}_2$ if and only if there exist $n = \text{mincut}(s, t)$ edge-disjoint paths $P_1, P_2, \dots, P_n$ from s to t such that $e_{1,i} \leq e_{2,i}$, where $P_i \cap \text{CUT}_1 = \{e_{1,i}\}$ and $P_i \cap \text{CUT}_2 = \{e_{2,i}\}$, $\forall 1 \leq i \leq n$.

Lemma 6 ([22, Lemma 15]): Let $n = \text{mincut}(s, t)$ and $\text{CUT}_1, \text{CUT}_2 \in \text{MinCut}(s, t)$ with $\text{CUT}_1 \leq \text{CUT}_2$. Consider an arbitrary set of n edge-disjoint paths $P_1, P_2, \dots, P_n$ from s to t . Let $P_i \cap \text{CUT}_1 = \{e_{1,i}\}$ and $P_i \cap \text{CUT}_2 = \{e_{2,i}\}$, $1 \leq i \leq n$. Then $e_{1,i} \leq e_{2,i}$ holds for all $i = 1, 2, \dots, n$.

Proof of Theorem 4: We first prove that $\text{CUT}_{\min}^{\mathcal{P}}$ and $\text{CUT}_{\max}^{\mathcal{P}}$ are two minimum cuts separating t from s . Note that it has been proved in [26, Lemma 5] that $\text{CUT}_{\min}^{\mathcal{P}}$ is a minimum cut separating t from s . It thus suffices to prove that $\text{CUT}_{\max}^{\mathcal{P}}$ is a minimum cut separating t from s . We consider the reverse graph $G^\top = (V, E^\top)$, where $E^\top = \{e^\top : e \in E\}$ and here $e^\top \in E^\top$ is the edge on $G^\top$ corresponding to edge e in E with reversed direction. Similarly, for an edge subset $\rho \subseteq E$ in G , we let $\rho^\top \triangleq \{e^\top : e \in \rho\} \subseteq E^\top$ be the edge subset on $G^\top$ associated with ρ . Following from the fact that for any two edges d and e in G , $d \leq e$ on G if and only if $e^\top \leq d^\top$ on $G^\top$, we readily see that for each $1 \leq i \leq n$, $\text{maxord}(e_{1,i}, e_{2,i})$ in E is the same edge as $\text{minord}(e_{1,i}^\top, e_{2,i}^\top)$ in $E^\top$ except for the direction, i.e.,

$$\text{maxord}(e_{1,i}, e_{2,i}) = \text{minord}(e_{1,i}^\top, e_{2,i}^\top)^\top, \quad (6)$$

where we recall that $\{e_{1,i}\} = P_i \cap \text{CUT}_1$ and $\{e_{2,i}\} = P_i \cap \text{CUT}_2$. Further, by the fact that $\text{CUT}_1^\top$ and $\text{CUT}_2^\top$ are two minimum

cuts separating s from t on the reverse graph $G^\top$, we claim that the edge subset

$$\{\text{minord}(e_{1,i}^\top, e_{2,i}^\top) : i = 1, 2, \dots, n\} \quad (7)$$

is also a minimum cut on $G^\top$ separating s from t . By (6) we can write

$$\begin{aligned} \text{CUT}_{\max}^{\mathcal{P}} &= \{\text{maxord}(e_{1,i}, e_{2,i}) : i = 1, 2, \dots, n\} \\ &= \{\text{minord}(e_{1,i}^\top, e_{2,i}^\top)^\top : i = 1, 2, \dots, n\} \\ &= \{\text{minord}(e_{1,i}^\top, e_{2,i}^\top) : i = 1, 2, \dots, n\}^\top, \end{aligned}$$

which, together with (7), shows that $\text{CUT}_{\max}^{\mathcal{P}}$ is a minimum cut separating t from s in G .

Next, we will prove (5) for any two sets of n edge-disjoint paths from s to t , say $\mathcal{P} = \{P_1, P_2, \dots, P_n\}$ and $\mathcal{P}' = \{P'_1, P'_2, \dots, P'_n\}$. By the above discussion, it suffices to prove $\text{CUT}_{\min}^{\mathcal{P}'} = \text{CUT}_{\min}^{\mathcal{P}}$. By Lemma 5 and the definitions of $\text{CUT}_{\min}^{\mathcal{P}}$ and $\text{CUT}_{\min}^{\mathcal{P}'}$, we have

$$\text{CUT}_{\min}^{\mathcal{P}} \leq \text{CUT}_i \quad \text{and} \quad \text{CUT}_{\min}^{\mathcal{P}'} \leq \text{CUT}_i, \quad i = 1, 2. \quad (8)$$

We assume the contrary that $\text{CUT}_{\min}^{\mathcal{P}} \neq \text{CUT}_{\min}^{\mathcal{P}'}$. Then, there exists an edge, say $e'_{1,1}$ (here $P'_1 \cap \text{CUT}_1 = \{e'_{1,1}\}$), such that $e'_{1,1} \in \text{CUT}_{\min}^{\mathcal{P}'}$ but $e'_{1,1} \notin \text{CUT}_{\min}^{\mathcal{P}}$. Following from Lemma 3 and the fact that CUT_1 is a minimum cut separating t from s , the edge $e'_{1,1} \in \text{CUT}_1$ must be on some path in $\mathcal{P}$, say P_2 , and then $e'_{1,1} = e_{1,2}$ where $P_2 \cap \text{CUT}_1 = \{e_{1,2}\}$. Furthermore, since $e_{1,2} (= e'_{1,1}) \notin \text{CUT}_{\min}^{\mathcal{P}}$, we immediately obtain that $e_{2,2} \in \text{CUT}_{\min}^{\mathcal{P}}$ (where $\{e_{2,2}\} = P_2 \cap \text{CUT}_2$) and $e_{2,2} \leq e_{1,2} = e'_{1,1}$ due to the definition of $\text{CUT}_{\min}^{\mathcal{P}}$ (cf. (2) and (3)). Thus, for the two minimum cuts $\text{CUT}_{\min}^{\mathcal{P}}$ and CUT_2 separating t from s and the n edge-disjoint paths $P_1, P_2, \dots, P_n$ from s to t , we have obtained that $\text{CUT}_{\min}^{\mathcal{P}'} \leq \text{CUT}_2$ by (8) and $e_{2,2} \leq e_{1,2}$, where $\{e_{2,2}\} = P_2 \cap \text{CUT}_2$ and $\{e_{1,2}\} = P_2 \cap \text{CUT}_{\min}^{\mathcal{P}'}$. This contradicts Lemma 6. Thus, we have proved $\text{CUT}_{\min}^{\mathcal{P}} = \text{CUT}_{\min}^{\mathcal{P}'}$ and also the theorem. ■

It follows from Theorem 4 that $\text{CUT}_{\min}^{\mathcal{P}}$ and $\text{CUT}_{\max}^{\mathcal{P}}$ (cf. (3) and (4)) are independent of which set of n edge-disjoint paths is chosen. Thus, in the sequel, it suffices to write $\text{CUT}_{\min}^{\mathcal{P}}$ and $\text{CUT}_{\max}^{\mathcal{P}}$, respectively, as

$$\text{CUT}_{\min} \triangleq \text{minord}(\text{CUT}_1, \text{CUT}_2)$$

and

$$\text{CUT}_{\max} \triangleq \text{maxord}(\text{CUT}_1, \text{CUT}_2),$$

which do not cause any ambiguity at all. A straightforward consequence of Theorem 4 is given as follows.

Corollary 7: Let CUT_1 and CUT_2 be two minimum cuts in $\text{MinCut}(s, t)$. Then,

$$\text{CUT}_{\min} \leq \text{CUT}_i \leq \text{CUT}_{\max}, \quad i = 1, 2,$$

where $\text{CUT}_{\min} = \text{minord}(\text{CUT}_1, \text{CUT}_2)$ and $\text{CUT}_{\max} = \text{maxord}(\text{CUT}_1, \text{CUT}_2)$.

We are now ready to prove Theorem 2.

Proof of Theorem 2: To show that $(\text{Cl}_r, \leq)$ is a meet-semilattice, we need to prove that for any two edge subsets ρ and η in Cl_r , the meet $\rho \wedge \eta$ exists, i.e., there exists the infimum

(greatest lower bound) of $\{\rho, \eta\}$. We first write up the set of lower bounds of $\{\rho, \eta\}$ as follows

$$\{\rho, \eta\}^\ell = \{\xi \in \text{Cl}_r : \xi \leq \rho \text{ and } \xi \leq \eta\}.$$

We note that $\{\rho, \eta\}^\ell$ is nonempty because $\rho \sim \eta$ so that ρ and η must have a common minimum cut separating each of them from s .

It now remains to prove the existence of the infimum of $\{\rho, \eta\}$, or equivalently, the uniqueness of the maximal elements of $\{\rho, \eta\}^\ell$. This will be proved by contradiction as follows. We assume the contrary that there exist two distinct maximal elements of $\{\rho, \eta\}^\ell$, denoted by ξ_1 and ξ_2 . This immediately implies that

$$\xi_1 \not\leq \xi_2 \quad \text{and} \quad \xi_2 \not\leq \xi_1. \quad (9)$$

For $i = 1, 2$, it is not difficult to see that ξ_i is a minimum cut separating $\rho \cup \eta$ from s . We consider subdividing each edge e in $\rho \cup \eta$ by introducing a node v_e and splitting e into two edges $e^{(1)}$ and $e^{(2)}$ such that $\text{tail}(e^{(1)}) = \text{tail}(e)$, $\text{head}(e^{(2)}) = \text{head}(e)$, and $\text{head}(e^{(1)}) = \text{tail}(e^{(2)}) = v_e$. Let $V_{\rho \cup \eta} = \{v_e : e \in \rho \cup \eta\}$. We further introduce a node $\hat{t}$ and add a “super-edge” of infinite capacity from v_e to $\hat{t}$ for each $v_e \in V_{\rho \cup \eta}$. By the definition of minimum cut separating an edge subset from a node in Section II, ξ_1 and ξ_2 , which are two minimum cuts separating $\rho \cup \eta$ from s , are minimum cuts separating $\hat{t}$ from s . By Theorem 4, the edge subset $\xi_{\max} \triangleq \text{maxord}(\xi_1, \xi_2)$ is also a minimum cut separating $\hat{t}$ from s and thus a minimum cut separating $\rho \cup \eta$ from s due to the definition. Together with $|\xi_{\max}| = r$, $\xi_{\max}$ is a minimum cut separating ρ from s and also a minimum cut separating η from s . By Corollary 7, we further have

$$\xi_1 \leq \xi_{\max} \quad \text{and} \quad \xi_2 \leq \xi_{\max}.$$

This implies that $\xi_{\max} \in \text{Cl}_r$ and by (9), $\xi_{\max} \neq \xi_i$ for $i = 1, 2$. We thus have proved that

$$\{\xi_1, \xi_2\} \leq \xi_{\max} \leq \{\rho, \eta\}.$$

Here, for two sets A and B , we use $A \leq B$ to denote $a \leq b$ for all $a \in A$ and $b \in B$. This contradicts to the assumption that ξ_1 and ξ_2 are two distinct maximal elements of $\{\rho, \eta\}^\ell$. Hence, we have already proved that $(\text{Cl}_r, \leq)$ is a meet-semilattice. ■

Similar to the definitions introduced for a source node, we now introduce the corresponding notions for a sink node. We say an edge subset ρ on the graph G is *regular to the sink node* t , or *regular- t* for simplicity, if $|\rho| = \text{mincut}(\rho, t)$, where $\text{mincut}(\rho, t)$ is the minimum cut capacity separating t from ρ . For two edge subsets ρ and η , we say ρ and η have a binary relation “ $\sim_t$ ”, denoted by $\rho \sim_t \eta$, if and only if ρ and η have a common minimum cut separating t from each of them. It is evident that if two edge subsets ρ and η are regular- t , then $\rho \sim_t \eta$ immediately implies $|\rho| = |\eta|$. For the graph $G = (V, E)$, we recall its reverse graph $G^\top = (V, E^\top)$, obtained by reversing the direction of every edge on G . We readily see that an edge subset ρ is regular to t on G if and only if $\rho^\top$ is regular from t on $G^\top$. Consequently, for two regular- t edge subsets ρ and η in G , $\rho \sim_t \eta$ on G if and only if $\rho^\top \sim_{t^\top} \eta^\top$ on $G^\top$. Thus, the binary relation “ $\sim_t$ ” on G is also an equivalence relation because “ $\sim_t$ ” on $G^\top$ is an equivalence relation. Similarly, we

consider a nonempty collection of all the regular- t edge subsets of size r

$$\mathcal{B}_t(r) = \{\rho \subseteq E : |\rho| = \text{mincut}(\rho, t) = r\}; \quad (10)$$

and define the binary relation “ $\leq_t$ ” among the edge subsets in $\mathcal{B}_t(r)$ as follows: for two regular- t edge subsets ρ and η in $\mathcal{B}_t(r)$, we write $\rho \leq_t \eta$ if no path exists from any edge in ρ to t upon deleting the edges in η . With the duality of the graph G and its reverse graph $G^\top$, we similarly have the following theorem.

Theorem 8: The binary relation “ $\leq_t$ ” on $\mathcal{B}_t(r)$ is a partial order. Let cl_r be an equivalence class of $\mathcal{B}_t(r)$. Then $(\text{cl}_r, \leq_t)$ is a join-semilattice.

We continue to use the graph G in Example 2 to illustrate the join-semilattices of edge subsets.

Example 3: Recall the graph $G = (V, E)$ depicted in Fig. 4. Let $r = 2$. For the sink node t_1 , we consider the collection of all regular- t_1 edge subsets of size 2, i.e.,

$$\mathcal{B}_{t_1}(2) = \{\rho \subseteq E : |\rho| = \text{mincut}(\rho, t_1) = 2\}$$

(cf. (10)). In the following, we present all the 17 equivalence classes of $\mathcal{B}_{t_1}(2)$:

$$\begin{aligned} \text{cl}_2^{(1)} &= \{\{e_1, e_4\}\}, & \text{cl}_2^{(2)} &= \{\{e_1, e_{10}\}, \{e_1, e_3\}\}, \\ \text{cl}_2^{(3)} &= \{\{e_1, e_{12}\}\}, \\ \text{cl}_2^{(4)} &= \{\{e_1, e_{20}\}, \{e_1, e_5\}, \{e_1, e_{13}\}, \{e_1, e_{14}\}, \{e_1, e_{17}\}\}, \\ \text{cl}_2^{(5)} &= \{\{e_4, e_6\}\}, & \text{cl}_2^{(6)} &= \{\{e_4, e_{10}\}, \{e_3, e_4\}\}, \\ \text{cl}_2^{(7)} &= \{\{e_4, e_{18}\}, \{e_2, e_4\}, \{e_4, e_7\}, \{e_4, e_8\}, \{e_4, e_{16}\}\}, \\ \text{cl}_2^{(8)} &= \{\{e_6, e_{10}\}, \{e_3, e_6\}\}, & \text{cl}_2^{(9)} &= \{\{e_6, e_{12}\}\}, \\ \text{cl}_2^{(10)} &= \{\{e_6, e_{18}\}, \{e_1, e_2\}, \{e_1, e_6\}, \{e_1, e_7\}, \{e_1, e_8\}, \\ & \quad \{e_1, e_{16}\}, \{e_1, e_{18}\}, \{e_2, e_6\}, \{e_6, e_7\}, \{e_6, e_8\}, \\ & \quad \{e_6, e_{16}\}\}, \\ \text{cl}_2^{(11)} &= \{\{e_6, e_{20}\}, \{e_5, e_6\}, \{e_6, e_{13}\}, \{e_6, e_{14}\}, \{e_6, e_{17}\}\}, \\ \text{cl}_2^{(12)} &= \{\{e_{10}, e_{12}\}, \{e_3, e_{12}\}\}, \\ \text{cl}_2^{(13)} &= \{\{e_{10}, e_{18}\}, \{e_2, e_3\}, \{e_2, e_{10}\}, \{e_3, e_7\}, \\ & \quad \{e_3, e_8\}, \{e_3, e_{16}\}, \{e_3, e_{18}\}, \{e_7, e_{10}\}, \\ & \quad \{e_8, e_{10}\}, \{e_{10}, e_{16}\}\}, \\ \text{cl}_2^{(14)} &= \{\{e_{10}, e_{20}\}, \{e_3, e_5\}, \{e_3, e_{13}\}, \{e_3, e_{14}\}, \\ & \quad \{e_3, e_{17}\}, \{e_3, e_{20}\}, \{e_5, e_{10}\}, \{e_{10}, e_{13}\}, \\ & \quad \{e_{10}, e_{14}\}, \{e_{10}, e_{17}\}\}, \\ \text{cl}_2^{(15)} &= \{\{e_{12}, e_{18}\}, \{e_2, e_{12}\}, \{e_7, e_{12}\}, \{e_8, e_{12}\}, \\ & \quad \{e_{12}, e_{16}\}\}, \\ \text{cl}_2^{(16)} &= \{\{e_{12}, e_{20}\}, \{e_4, e_5\}, \{e_4, e_{12}\}, \{e_4, e_{13}\}, \\ & \quad \{e_4, e_{14}\}, \{e_4, e_{17}\}, \{e_4, e_{20}\}, \{e_5, e_{12}\}, \\ & \quad \{e_{12}, e_{13}\}, \{e_{12}, e_{14}\}, \{e_{12}, e_{17}\}\}, \\ \text{cl}_2^{(17)} &= \{\{e_{18}, e_{20}\}, \{e_2, e_5\}, \{e_2, e_{13}\}, \{e_2, e_{14}\}, \\ & \quad \{e_2, e_{17}\}, \{e_2, e_{20}\}, \{e_5, e_7\}, \{e_5, e_8\}, \\ & \quad \{e_5, e_{16}\}, \{e_5, e_{18}\}, \{e_7, e_{13}\}, \{e_7, e_{14}\}, \\ & \quad \{e_7, e_{17}\}, \{e_7, e_{20}\}, \{e_8, e_{13}\}, \{e_8, e_{14}\}, \\ & \quad \{e_8, e_{17}\}, \{e_8, e_{20}\}, \{e_{13}, e_{16}\}, \{e_{13}, e_{18}\}\}, \end{aligned}$$

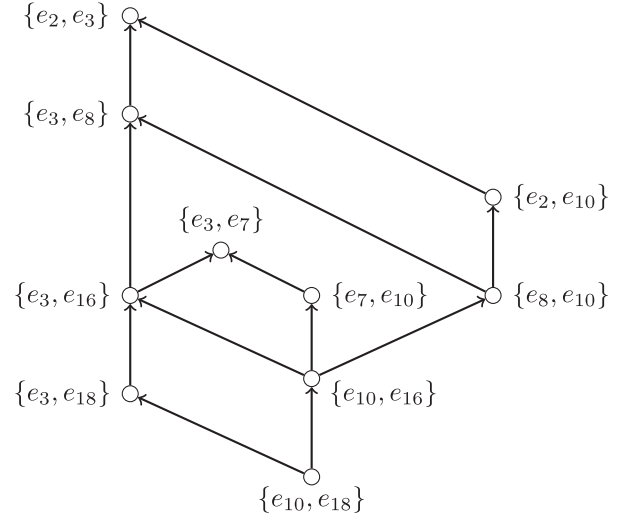


Fig. 5. The Hasse diagram of the join-semilattice $(\text{cl}_2^{(13)}, \leq_{t_1})$.

$$\{e_{14}, e_{16}\}, \{e_{14}, e_{18}\}, \{e_{16}, e_{17}\}, \{e_{16}, e_{20}\}, \\ \{e_{17}, e_{18}\}.$$

By Theorem 8, each equivalence class partially ordered by the binary relation “ $\leq_t$ ” is a join-semilattice, e.g., $(\text{cl}_2^{(13)}, \leq_{t_1})$. The Hasse diagram of this join-semilattice $(\text{cl}_2^{(13)}, \leq_{t_1})$ is illustrated in Fig. 5, where the nodes represent all the 10 size-2 regular- t_1 edge subsets in $\text{cl}_2^{(13)}$. For any two edge subsets ρ and η in $\text{cl}_2^{(13)}$, we say ρ is covered by η (or η covers ρ), written as $\rho \leftarrow \eta$ (or $\eta \rightarrow \rho$), if i) $\rho \leq_{t_1} \eta$ and $\rho \neq \eta$, and ii) for any edge subset $\xi \in \text{cl}_2^{(13)}$ with $\rho \leq_{t_1} \xi \leq_{t_1} \eta$, either $\rho = \xi$ or $\eta = \xi$. For two nodes associated with two edge subsets in $\text{cl}_2^{(13)}$, say ρ and η , there exists a directed edge from η to ρ if η covers ρ . With the Hasse diagram in Fig. 5, we can see that the edge subset $\{e_{10}, e_{18}\}$ is the top element of the join-semilattice $(\text{cl}_2^{(13)}, \leq_{t_1})$, which separates t_1 from all the edge subsets in $\text{cl}_2^{(13)}$.

Next, we consider the minimum cuts separating t from s . We first present a similar corollary to Corollary 7, which is also a straightforward consequence of Theorem 4.

Corollary 9: Let CUT_1 and CUT_2 be two minimum cuts in $\text{MinCut}(s, t)$, and further let

$$\begin{aligned} \text{CUT}_{\min} &= \text{minord}(\text{CUT}_1, \text{CUT}_2), \\ \text{CUT}_{\max} &= \text{maxord}(\text{CUT}_1, \text{CUT}_2). \end{aligned}$$

Then,

$$\text{CUT}_{\min} \leq_t \text{CUT}_i \leq_t \text{CUT}_{\max}, \quad i = 1, 2.$$

We further obtain the corollary below by the above discussions.

Corollary 10: Let CUT_1 and CUT_2 be two minimum cuts in $\text{MinCut}(s, t)$. Then,

- 1) $\text{CUT}_1 \sim_s \text{CUT}_2$ and $\text{CUT}_1 \sim_t \text{CUT}_2$;
- 2) $\text{CUT}_1 \leq_s \text{CUT}_2$ if and only if $\text{CUT}_1 \leq_t \text{CUT}_2$, which implies that the binary relations “ $\leq_s$ ” and “ $\leq_t$ ” are identical. Hence, we use “ $\leq$ ” here to unify “ $\leq_s$ ” and “ $\leq_t$ ”;
- 3) $\text{CUT}_{\min} \leq \text{CUT}_i \leq \text{CUT}_{\max}$ for $i = 1, 2$.

Proof: For the two minimum cuts CUT_1 and CUT_2 in $\text{MinCut}(s, t)$, let $CUT_{\min} = \text{minord}(CUT_1, CUT_2)$ and $CUT_{\max} = \text{maxord}(CUT_1, CUT_2)$. It now follows from Corollaries 7 and 9 that $CUT_{\min}$ is a common minimum cut separating CUT_1 and CUT_2 from s , and $CUT_{\max}$ is a common minimum cut separating t from CUT_1 and CUT_2 . Together with the definitions of “ $\sim$ ” and “ $\sim_t$ ”, these immediately imply that $CUT_1 \sim CUT_2$ and $CUT_1 \sim_t CUT_2$.

Next, we prove that $CUT_1 \leq_s CUT_2$ if and only if $CUT_1 \leq_t CUT_2$. First, we assume that $CUT_1 \leq_s CUT_2$ but $CUT_1 \not\leq_t CUT_2$. Since $CUT_1 \not\leq_t CUT_2$, namely that CUT_2 is not a minimum cut separating t from CUT_1 , there exists a path from an edge, say e , in CUT_1 to t without passing through CUT_2 . We denoted this path by $P^{e \rightarrow t}$. On the other hand, since $CUT_1 \leq_s CUT_2$, by Lemma 5 there exists a path from s to e without passing through CUT_2 , denoted by $P^{s \rightarrow e}$. We now concatenate $P^{s \rightarrow e}$ and $P^{e \rightarrow t}$ to obtain a path from s to t without passing through CUT_2 , which contradicts the fact that CUT_2 is a minimum cut separating t from s . We thus have proved that $CUT_1 \leq_s CUT_2$ implies $CUT_1 \leq_t CUT_2$. By using the same argument, we can further prove that $CUT_1 \leq_t CUT_2$ implies $CUT_1 \leq_s CUT_2$.

It immediately follows from 2) and Corollary 7 (or Corollary 9) that

$$CUT_{\min} \leq CUT_i \leq CUT_{\max} \quad \text{for } i = 1, 2.$$

The corollary is proved. ■

Combining Theorems 2 and 8 and Corollary 10, we readily have the theorem below which asserts that $(\text{MinCut}(s, t), \leq)$ is a lattice.

Theorem 11: The partially ordered set $(\text{MinCut}(s, t), \leq)$ is a lattice. Further, for any two minimum cuts CUT_1 and CUT_2 in $\text{MinCut}(s, t)$,

$$CUT_1 \wedge CUT_2 = \text{minord}(CUT_1, CUT_2)$$

and

$$CUT_1 \vee CUT_2 = \text{maxord}(CUT_1, CUT_2).$$

Proof: By Theorems 2 and 8, Corollary 10 and the definition of lattice (cf. Definition 2), we readily see that $(\text{MinCut}(s, t), \leq)$ is a lattice. Next, we consider two minimum cuts CUT_1 and CUT_2 in $\text{MinCut}(s, t)$. Let $CUT_{\min} = \text{minord}(CUT_1, CUT_2)$. It follows from Corollary 10 that $CUT_{\min} \leq CUT_i$, $i = 1, 2$, namely that $CUT_{\min} \in \{CUT_1, CUT_2\}^\ell$. We will prove that $CUT_{\min}$ is the supremum (greatest lower bound) of CUT_1 and CUT_2 . For any lower bound $\xi \in \{CUT_1, CUT_2\}^\ell$, $\xi \leq CUT_1$ and $\xi \leq CUT_2$ implies that ξ separates $CUT_1 \cup CUT_2$ from s . Together with $CUT_{\min} \subseteq CUT_1 \cup CUT_2$, ξ separates $CUT_{\min}$ from s and thus $\xi \leq CUT_{\min}$. This immediately shows that $CUT_{\min}$ is the supremum of CUT_1 and CUT_2 , i.e.,

$$CUT_1 \wedge CUT_2 = CUT_{\min} = \text{minord}(CUT_1, CUT_2).$$

By using the same argument, we can similarly prove that $CUT_1 \vee CUT_2 = \text{maxord}(CUT_1, CUT_2)$. The theorem is thus proved. ■

By Theorem 11, we immediately obtain the corollary below.

Corollary 12: For the lattice $(\text{MinCut}(s, t), \leq)$, there exists a unique minimum cut separating the sink node t from the

source node s if and only if the top $\top$ and the bottom $\perp$ are identical, i.e., $\top = \perp$.

By Proposition 2 and Theorem 2, the meet-semilattice $(Cl_r, \leq_s)$ has a (unique) bottom $\perp$, i.e., $\perp \in Cl_r$ and $\perp \leq_s \rho$ for all $\rho \in Cl_r$. In other words, for every edge subset ρ in the equivalence class Cl_r in which all the size- r s -regular edge subsets are equivalent, there exists no path from s to any edge in ρ upon deleting the edges in $\perp$, or equivalently, the bottom $\perp$ is the common minimum cut separating all the edge subsets in Cl_r from s . On the other hand, Guang and Yeung [22] introduced the concept of the *primary minimum cut* separating an edge subset ρ from s , which is defined as a minimum cut separating ρ from s that separates from s all the minimum cuts that separate ρ from s . The existence and uniqueness of the primary minimum cut were also proved in [22]. By comparison, we obtain an interesting discovery that for each s -regular edge subset $\rho \in Cl_r$, the primary minimum cut separating ρ from s and the bottom $\perp$ of the meet-semilattice $(Cl_r, \leq_s)$ are in fact identical. This discovery is stated formally in Theorem 13 below, which provides a novel way to understand the primary minimum cut and its existence and uniqueness.

Theorem 13: For an s -regular edge subset ρ , the primary minimum cut separating ρ from s is the bottom of the meet-semilattice $(Cl(\rho), \leq_s)$, where $Cl(\rho)$ is the equivalence class containing ρ , i.e., the set of the s -regular edge subsets that are equivalent to ρ under the equivalence relation “ $\sim_s$ ”.

Also, we can obtain a similar theorem for regular- t edge subsets. We first define the *primary minimum cut* separating t from an edge subset ρ by the minimum cut separating t from ρ that separates t from all the minimum cuts separating t from ρ .

Theorem 14: For a regular- t edge subset ρ , the primary minimum cut separating t from ρ is the top of the join-semilattice $(cl(\rho), \leq_t)$, where $cl(\rho)$ is the equivalence class containing ρ , i.e., the set of the regular- t edge subsets that are equivalent to ρ under the equivalence relation “ $\sim_t$ ”.

Further, we consider the lattice $(\text{MinCut}(s, t), \leq)$. Together with Proposition 2 and Theorems 13 and 14, the bottom $\perp$ is the minimum cut in $\text{MinCut}(s, t)$ that separates all the minimum cuts in $\text{MinCut}(s, t)$ from s , which is the minimum cut “nearest” to the source node s ; and the top $\top$ is the minimum cut in $\text{MinCut}(s, t)$ that separates t from all the minimum cuts in $\text{MinCut}(s, t)$, which is the minimum cut “nearest” to the sink node t . This enhances the concept of the primary minimum cut between s and t introduced in [22], which is defined by a minimum cut separating t from s that separates all the minimum cuts in $\text{MinCut}(s, t)$ from s . This primary minimum cut is indeed the bottom $\perp$ in $\text{MinCut}(s, t)$, i.e., the minimum cut “nearest” to the source node s . However, the top $\top$ in $\text{MinCut}(s, t)$, namely the minimum cut “nearest” to the sink node t , escaped from the attention in [22].

C. Properties of Edge-Subset Semilattice

In this subsection, we discuss some properties of edge-subset semilattices thus obtained, which are of importance to improve the required field size for the existence of LNEC codes in the sequel. We continue to consider the graph G with

the single source node s and the sink node t . Let r be a positive integer such that $\mathcal{B}_t(r)$ is nonempty. We let $\mathcal{A}_t(r)$ be the set of the top elements of the join-semilattices $(\text{cl}_r, \leq_t)$ for all the equivalence classes cl_r of $\mathcal{B}_t(r)$, i.e.,

$$\mathcal{A}_t(r) = \left\{ \rho \subseteq E : \rho \text{ is the top of } (\text{cl}_r, \leq_t) \text{ for an equivalence class } \text{cl}_r \text{ of } \mathcal{B}_t(r) \right\}.^2 \quad (11)$$

On the other hand, in [9], an edge subset ρ is said to be *primary for t* if ρ itself is the primary minimum cut separating t from ρ . Then, we can see that a size- r primary edge subset for t is a top element in $\mathcal{A}_t(r)$, and vice versa. Accordingly, we can further write $\mathcal{A}_t(r)$ as

$$\mathcal{A}_t(r) = \left\{ \rho \subseteq E : \rho \text{ is a size-} r \text{ primary edge subset for } t \right\}. \quad (12)$$

Next, we present the following property on primary minimum cuts (or top elements).

Proposition 3: Let $\text{SL}_t(r)$ be the set of all the join-semilattices $(\text{cl}_r, \leq_t)$ for the equivalence classes cl_r of $\mathcal{B}_t(r)$, i.e.,

$$\text{SL}_t(r) = \{ (\text{cl}_r, \leq_t) : \text{for all equivalence classes } \text{cl}_r \text{ of } \mathcal{B}_t(r) \}.$$

Let η be the top of a join-semilattice in $\text{SL}_t(r)$. Consider an arbitrary nonempty subset ρ of η . Then ρ is a top for some join-semilattice in $\text{SL}_t(|\rho|)$. In other words, a subset of a primary edge set for t is also primary for t .

Proof: We will prove this proposition by contradiction. Without loss of generality, we assume that ρ is not primary for t . Then, let $\tilde{\rho} \subseteq E$ be the primary minimum cut separating t from ρ , and $\tilde{\rho} \neq \rho$. Consider an edge subset $\tilde{\eta} = \tilde{\rho} \cup (\eta \setminus \rho)$. Clearly, we have $\tilde{\eta} \neq \eta$.

We now prove that $\tilde{\eta}$ is a cut separating t from η . First, since $\eta \setminus \rho \subset \tilde{\eta}$, we readily see that upon deleting the edges in $\tilde{\eta}$, no path exists from any edge e in $\eta \setminus \rho$ to t . Further, since $\tilde{\rho}$ is a minimum cut separating t from ρ , together with $\tilde{\rho} \subseteq \tilde{\eta}$, no path exists from any edge $e \in \rho$ to t upon deleting the edges in $\tilde{\eta}$, either. Thus, we have proved that $\tilde{\eta}$ is a cut separating t from η .

Furthermore, we obtain that

$$|\tilde{\eta}| = |\tilde{\rho} \cup (\eta \setminus \rho)| \leq |\tilde{\rho}| + |\eta \setminus \rho| \leq |\rho| + |\eta \setminus \rho| = |\eta|.$$

Thus, we have a cut $\tilde{\eta}$ separating t from η such that $\tilde{\eta} \neq \eta$ and $|\tilde{\eta}| \leq |\eta|$. This contradicts the assumption that η is primary for t . The proposition is thus proved. ■

We immediately obtain the following consequence of Proposition 3.

Corollary 15: Let $\eta = \{e_1, e_2, \dots, e_r\}$ be a size- r primary edge set for the sink node t . Then, the singleton edge subset $\{e_i\}$ for each $1 \leq i \leq r$ is also primary for t .

We remark that the converse of Corollary 15 (also Proposition 3) is not true, namely that the edge set $\eta = \{e_1, e_2, \dots, e_r\}$ may be not primary for t even if each of the singleton edge subsets $\{e_i\}$, $1 \leq i \leq r$ is primary for t . We give a

counterexample as follows. We continue to consider the graph G depicted in Fig. 4. Note that $\{e_1\}$ and $\{e_6\}$ are two size-1 primary edge subsets for t_1 , but $\{e_1, e_6\}$ is not primary for t_1 because the edge subset $\{e_6, e_{18}\}$ is the primary minimum cut separating t_1 from $\{e_1, e_6\}$.

Next, we consider an interesting and important problem of how to find the top and/or the bottom of an edge-subset (semi-)lattice, because the top and the bottom here are graph-theoretic so that they are not given in a form which is readily computable. Hence, the best one can ever hope for is to develop efficient algorithms to find them. We recall the discussions in the paragraph below Corollary 12 that for a size- r s -regular edge subset ρ , the bottom $\perp$ of the meet-semilattice $(\text{Cl}(\rho), \leq_s)$ is the primary minimum cut separating ρ from s , where $\text{Cl}(\rho)$ is the equivalence class of $\mathcal{B}_s(r)$ that contains ρ (cf. (1) for $\mathcal{B}_s(r)$). Guang and Yeung in [22] have developed an efficient algorithm to find primary minimum cut (see Algorithm 6 in [22]). Thus, by modifying this algorithm, we can develop efficient algorithms to find the bottom of a meet-semilattice and the top of a join-semilattice. For the completeness of the paper, in Appendix B we present the two algorithms to find the bottom and the top, respectively, followed by an example to illustrate the two algorithms and the complexity analysis of the algorithms.

IV. FIELD SIZE REDUCTION FOR LNEC CODES

The minimum required field size for the existence of LNEC codes, in particular LNEC maximum distance separable (MDS) codes which are regarded as a most important type of optimal codes, is an open problem not only of theoretical interest but also of practical importance, because it is closely related to the computational complexity and storage requirement of the construction and implementation of a code. Based on the discussions of edge-subset (semi-)lattices in the last section, we will present an improved upper bound on the minimum required field size, which shows that the required field size for the existence of LNEC codes in general can be reduced significantly. This new bound is graph-theoretic, which depends only on the network topology and requirement of error correction capability but not on the specific code construction. Before presenting the results, we first revisit LNEC coding.

A. Framework of Additive Adversarial Network for LNEC Coding

In this subsection, we briefly present the framework of *additive adversarial network* of LNEC coding [9]. We refer the reader to [9] for more details. Consider a communication network $G = (V, E)$, where the source node s is required to multicast the source message to all the sink nodes in T , which decode the source message while correcting errors occurring on some of the edges. In LNEC coding over a finite field $\mathbb{F}_q$ (q a prime power), all the symbols in the network, including the symbols generated by the source node s , the symbols transmitted on the edges and the errors occurred on the edges, are elements in $\mathbb{F}_q$. Let the *source message* $\mathbf{x} = (x_1 \ x_2 \ \dots \ x_\omega) \in \mathbb{F}_q^\omega$ be the row vector of the ω source

²Note that $\mathcal{A}_t(r) = \emptyset$ when $r = 0$.

symbols generated by the source node s , where ω is the (information) rate of the source. An error $z_e \in \mathbb{F}_q$ may occur on an edge $e \in E$.³ We write all the errors on the edges in E as an $\mathbb{F}_q$ -valued row $|E|$ -vector $\mathbf{z} = (z_e : e \in E)$, called the *error vector*.

For a source message vector $\mathbf{x} \in \mathbb{F}_q^\omega$ and an error vector $\mathbf{z} \in \mathbb{F}_q^{|E|}$, we denote by $\tilde{\mathbf{y}}_e(\mathbf{x}, \mathbf{z})$ the symbol received on an edge e . For a sink node $t \in T$, let $\tilde{\mathbf{y}}_t(\mathbf{x}, \mathbf{z}) \triangleq (\tilde{\mathbf{y}}_e(\mathbf{x}, \mathbf{z}) : e \in \text{In}(t))$. For an additive adversarial network of LNEC coding, we have

$$\tilde{\mathbf{y}}_t(\mathbf{x}, \mathbf{z}) = \mathbf{x} \cdot \mathbf{F}_t + \mathbf{z} \cdot \mathbf{G}_t, \quad (13)$$

where $\mathbf{F}_t$ and $\mathbf{G}_t$ are two $\mathbb{F}_q$ -valued matrices of sizes $\omega \times |\text{In}(t)|$ and $|E| \times |\text{In}(t)|$, respectively (cf. [9] for details). When $\mathbf{x}$ and $\mathbf{z}$ are clear from the context, we write $\tilde{\mathbf{y}}_e$ and $\tilde{\mathbf{y}}_t$ to simplify the notations. In (13), the matrices $\mathbf{F}_t$ and $\mathbf{G}_t$ and the vector $\tilde{\mathbf{y}}_t \in \mathbb{F}_q^{|\text{In}(t)|}$ are known to the receiver at the sink node t while $\mathbf{x}$ and $\mathbf{z}$ are unknown. Here, we always assume that $\text{Rank}(\mathbf{F}_t) = \omega$ for all $t \in T$ to guarantee that all the sink nodes in T can decode the source message successfully when no error occurs on the network.

For an LNEC code as described above, the set

$$\Phi(t) \triangleq \{\mathbf{x} \cdot \mathbf{F}_t : \text{all } \mathbf{x} \in \mathbb{F}_q^\omega\} \quad (14)$$

can be regarded as the “codebook” as seen at t , and is referred to as the *received codebook* at t . For an error vector $\mathbf{z} = (z_e : e \in E) \in \mathbb{F}_q^{|E|}$ and an edge subset $\rho \subseteq E$, we say that $\mathbf{z}$ *matches* ρ if $z_e = 0$ for all $e \in E \setminus \rho$. The error space of ρ at a sink node $t \in T$ is defined by

$$\Delta(t, \rho) = \{\mathbf{z} \cdot \mathbf{G}_t : \text{all } \mathbf{z} \in \mathbb{F}_q^{|E|} \text{ s.t. } \mathbf{z} \text{ matches } \rho\}. \quad (15)$$

It has been proved in [9] (also cf. [7], [10]) that the minimum distance $d_{\min}^{(t)}$ of the received codebook $\Phi(t)$ at t can be written as

$$d_{\min}^{(t)} = \min \{|\rho| : \Phi(t) \cap \Delta(t, \rho) \neq \{\mathbf{0}\}\}. \quad (16)$$

The minimum distance $d_{\min}^{(t)}$ characterizes both the error detection and error correction capability at the sink node $t \in T$ for the rate- ω LNEC code.

As a network generalization of the classical Singleton bound for a point-to-point error-correcting code, the (refined) *Singleton bound* for a sink node $t \in T$ is given by

$$d_{\min}^{(t)} \leq C_t - \omega + 1, \quad (17)$$

where we use C_t to denote the minimum cut capacity separating t from s for notational simplicity, i.e., $C_t = \text{mincut}(s, t)$. If an $\mathbb{F}_q$ -valued rate- ω LNEC code not only is decodable for T but also satisfies the Singleton bound (17) with equality for each sink node $t \in T$, i.e.,

$$\dim(\Phi(t)) = \omega \text{ and } d_{\min}^{(t)} = C_t - \omega + 1, \quad \forall t \in T,$$

then this code is called *maximum distance separable (MDS)* for T . An $\mathbb{F}_q$ -valued rate- ω LNEC MDS code achieves the maximum error correction capability at each sink node.

B. Improved Upper Bound on the Minimum Required Field Size

We first present our improved upper bound on the minimum required field size for the existence of LNEC codes in the following theorem.

Theorem 16: Let $\mathbb{F}_q$ be a finite field of order q . Let T be the set of sink nodes on the network G with $C_t \geq \omega$, $\forall t \in T$, where ω is the rate of the source. For each $t \in T$, let β_t be a nonnegative integer not larger than $C_t - \omega$. Then, there exists an $\mathbb{F}_q$ -valued rate- ω LNEC code on G with the minimum distance at t not smaller than $\beta_t + 1$ for each $t \in T$, i.e., $d_{\min}^{(t)} \geq \beta_t + 1$, $\forall t \in T$, if the field size q satisfies

$$q > \sum_{t \in T} |\mathcal{A}_t(\beta_t)| \quad (18)$$

(cf. (11) or (12) for $\mathcal{A}_t(\beta_t)$).

We first state a crucial lemma which was obtained in [9].

Lemma 17 ([9, Corollary 1]): Consider a rate- ω LNEC code over a finite field $\mathbb{F}_q$ on the network G . Then for a sink node t with $C_t \geq \omega$ and a nonnegative integer $r \leq C_t - \omega$, the following conditions are both equivalent:

$$\Phi(t) \cap \Delta(t, \rho) = \{\mathbf{0}\}, \quad \forall \rho \in \mathcal{A}_t(r),$$

and

$$\begin{aligned} \Phi(t) \cap \Delta(t, \rho) &= \{\mathbf{0}\}, \\ \forall \rho \in \mathcal{H}(r) &\triangleq \{\rho \subseteq E : |\rho| \leq r\}. \end{aligned} \quad (19)$$

Proof of Theorem 16: To prove this theorem, we need to prove that if (18) is satisfied for the field $\mathbb{F}_q$, then there exists an $\mathbb{F}_q$ -valued rate- ω LNEC code on G such that for each sink node $t \in T$,

$$\dim(\Phi(t)) = \omega \quad \text{and} \quad d_{\min}^{(t)} \geq \beta_t + 1. \quad (20)$$

With (16), for each $t \in T$, the condition (20) is equivalent to the following:

$$\begin{aligned} \dim(\Phi(t)) &= \omega, \quad \text{and} \\ \Phi(t) \cap \Delta(t, \rho) &= \{\mathbf{0}\}, \quad \forall \rho \subseteq E \text{ with } |\rho| \leq \beta_t. \end{aligned} \quad (21)$$

We further write the second constraint in (21) as

$$\Phi(t) \cap \Delta(t, \rho) = \{\mathbf{0}\}, \quad \forall \rho \in \mathcal{H}(\beta_t),$$

which, by Lemma 17, is equivalent to

$$\Phi(t) \cap \Delta(t, \rho) = \{\mathbf{0}\}, \quad \forall \rho \in \mathcal{A}_t(\beta_t).$$

Hence, the condition (20) is further equivalent to

$$\begin{aligned} \dim(\Phi(t)) &= \omega, \quad \text{and} \\ \Phi(t) \cap \Delta(t, \rho) &= \{\mathbf{0}\}, \quad \forall \rho \in \mathcal{A}_t(\beta_t). \end{aligned} \quad (22)$$

Based on the above discussion, it suffices to prove that if the field $\mathbb{F}_q$ satisfies (18), then there exists an $\mathbb{F}_q$ -valued rate- ω LNEC code on G such that the condition (22) is satisfied for each sink node $t \in T$. This statement can be proved by using a standard argument (e.g., the proof of Theorem 1 in [7] and the proof of Theorem 5 in [10]). We thus omit the details here. ■

A straightforward upper bound on the minimum required field size for the existence of a rate- ω LNEC code with the

³If no error occurs on the edge e , then $z_e = 0$.

minimum distance $d_{\min}^{(t)} \geq \beta_t + 1$ for each $t \in T$ (where β_t continues to denote a nonnegative integer not larger than $C_t - \omega$) is

$$\sum_{t \in T} |\mathcal{H}(\beta_t)| = \sum_{t \in T} \binom{|E|}{\beta_t}.$$

Such a code can correct at t an arbitrary error vector in the set

$$\mathcal{Z}(\mathcal{C}_t(\lfloor \beta_t/2 \rfloor)) \triangleq \{ \mathbf{z} \in \mathbb{F}_q^{|E|} : \mathbf{z} \text{ matches an edge subset } \rho \text{ with } \text{mincut}(\rho, t) \leq \lfloor \beta_t/2 \rfloor \}$$

for each $t \in T$ (cf. [9, Theorem 3]). Subsequently, this upper bound was improved in [10], as presented in the following proposition. To our knowledge, this is the best known upper bound on the minimum required field size for the existence of such a rate- ω LNEC code.

Proposition 4 ([10, Theorem 8]): Let $\mathbb{F}_q$ be a finite field of order q . Let T be the set of sink nodes on the network G with $C_t \geq \omega$, $\forall t \in T$, where ω is the rate of the source. For each $t \in T$, let β_t be a nonnegative integer not larger than $C_t - \omega$. Then, there exists an $\mathbb{F}_q$ -valued rate- ω LNEC code on G with the minimum distance $d_{\min}^{(t)} \geq \beta_t + 1$ for each $t \in T$ if the field size q satisfies

$$q > \sum_{t \in T} |\mathcal{B}_t(\beta_t)|, \quad (23)$$

where $\mathcal{B}_t(r) = \{ \rho \subseteq E : |\rho| = \text{mincut}(\rho, t) = r \}$ (cf. (10)).

We see that the collection $\mathcal{B}_t(\beta_t)$ is the union of all equivalence classes cl of $\mathcal{B}_t(\beta_t)$, i.e.,

$$\mathcal{B}_t(\beta_t) = \bigcup_{\text{all the equivalence classes cl}} \text{cl}. \quad (24)$$

Hence, we readily see that

$$\mathcal{A}_t(\beta_t) \subseteq \mathcal{B}_t(\beta_t) \subseteq \{ \rho \subseteq E : |\rho| = \beta_t \},$$

and their sizes are

$$|\mathcal{A}_t(\beta_t)| = \sum_{\text{all the equivalence classes cl}} 1, \quad |\mathcal{B}_t(\beta_t)| = \sum_{\text{all the equivalence classes cl}} |\text{cl}|,$$

and

$$|\{ \rho \subseteq E : |\rho| = \beta_t \}| = \binom{|E|}{\beta_t}.$$

This immediately shows that

$$\sum_{t \in T} |\mathcal{A}_t(\beta_t)| \leq \sum_{t \in T} |\mathcal{B}_t(\beta_t)| \leq \sum_{t \in T} \binom{|E|}{\beta_t},$$

and our obtained bound is smaller than the previous.

The improvement of our improved bound $\sum_{t \in T} |\mathcal{A}_t(\beta_t)|$ in Theorem 16 over $\sum_{t \in T} |\mathcal{B}_t(\beta_t)|$ (also over $\sum_{t \in T} \binom{|E|}{\beta_t}$) is in general significant, and Example 4 below shows this significance. The only case when $\sum_{t \in T} |\mathcal{A}_t(\beta_t)|$ has no improvement over $\sum_{t \in T} |\mathcal{B}_t(\beta_t)|$, i.e., $\sum_{t \in T} |\mathcal{A}_t(\beta_t)| = \sum_{t \in T} |\mathcal{B}_t(\beta_t)|$, is that each equivalence class cl only contains one element (i.e., its top). This condition holds only for very special networks. For example, consider a network that consists of only a source node s and a sink node t , connected by multiple parallel edges

from s to t (as depicted in Fig. 1). In the network, for any positive integer β_t not larger than $|E|$ (where $|E|$ is the number of multiple parallel edges from s to t), we readily see that each edge subset $\rho \subseteq E$ of size β_t itself forms an equivalence class. This immediately implies that

$$|\mathcal{A}_t(\beta_t)| = |\mathcal{B}_t(\beta_t)| = \binom{|E|}{\beta_t}, \quad \forall \beta_t \leq |E|.$$

Example 4: Recall the network $G = (V, E)$ depicted in Fig. 4, where s is the single source node and $T = \{t_1, t_2\}$ is the set of sink nodes with $C_{t_1} = C_{t_2} = 5$. Let the rate $\omega = 3$ and $\beta_{t_1} = \beta_{t_2} = 2$, two nonnegative integers not larger than $C_{t_1} - \omega$ and $C_{t_2} - \omega$, respectively. We consider an $\mathbb{F}_q$ -valued rate-3 LNEC code with $d_{\min}^{(t_1)} \geq \beta_{t_1} + 1$ and $d_{\min}^{(t_2)} \geq \beta_{t_2} + 1$. This code can correct at the sink node t_i an arbitrary error vector in the set $\mathcal{Z}(\mathcal{C}_{t_i}(1))$ for $i = 1, 2$. We now focus on the field size q for the existence of such a code.

We first calculate the straightforward bound $\sum_{t \in T} \binom{|E|}{\beta_t}$ on the field size q as follows:

$$\sum_{t \in T} \binom{|E|}{\beta_t} = 2 \cdot \binom{21}{2} = 420. \quad (25)$$

Next, we calculate the bound $\sum_{t \in T} |\mathcal{B}_t(\beta_t)|$ on the field size q in Proposition 4. With $\beta_{t_1} = 2$, we recall all the 17 equivalence classes of $\mathcal{B}_{t_1}(2)$ in Example 3, and then $\mathcal{B}_{t_1}(2) = \bigcup_{i=1}^{17} \text{cl}_i^{(t_1)}$ (cf. (24)). This immediately shows that $|\mathcal{B}_{t_1}(2)| = \sum_{i=1}^{17} |\text{cl}_i^{(t_1)}| = 99$. By the symmetry of the network G , we also have $|\mathcal{B}_{t_2}(2)| = 99$. So, the bound (23) in Proposition 4 is

$$|\mathcal{B}_{t_1}(2)| + |\mathcal{B}_{t_2}(2)| = 198, \quad (26)$$

which is smaller than 420 from (25).

Next, we present the set $\mathcal{A}_{t_1}(2)$ of all the tops of the equivalence classes of $\mathcal{B}_{t_1}(2)$, or equivalently, all the size-2 primary edge subsets for t_1 , as follows:

$$\begin{aligned} \mathcal{A}_{t_1}(2) = \{ & \{e_1, e_4\}, \{e_1, e_{10}\}, \{e_1, e_{12}\}, \{e_1, e_{20}\}, \\ & \{e_4, e_6\}, \{e_4, e_{10}\}, \{e_4, e_{18}\}, \{e_6, e_{10}\}, \\ & \{e_6, e_{12}\}, \{e_6, e_{18}\}, \{e_6, e_{20}\}, \{e_{10}, e_{12}\}, \\ & \{e_{10}, e_{18}\}, \{e_{10}, e_{20}\}, \{e_{12}, e_{18}\}, \{e_{12}, e_{20}\}, \\ & \{e_{18}, e_{20}\} \}. \end{aligned}$$

where we see $|\mathcal{A}_{t_1}(2)| = 17$, and for $1 \leq i \leq 17$, the i -th edge subset in $\mathcal{A}_{t_1}(2)$ is the top of cl_i , e.g., $\{e_1, e_4\}$ is the top of cl_1 . By the symmetry of G , we have $|\mathcal{A}_{t_2}(2)| = |\mathcal{A}_{t_1}(2)| = 17$, and thus the improved bound (18) in Theorem 16 is

$$|\mathcal{A}_{t_2}(2)| + |\mathcal{A}_{t_1}(2)| = 34,$$

which is considerably smaller than 198 from (26).

Furthermore, with the aid of a computer, we compute all $|\mathcal{A}_{t_1}(\beta_{t_1})|$ ($= |\mathcal{A}_{t_2}(\beta_{t_2})|$) and $|\mathcal{B}_{t_1}(\beta_{t_1})|$ ($= |\mathcal{B}_{t_2}(\beta_{t_2})|$) for $\beta_{t_1} = 3, 4, 5$ as follows⁴:

$$|\mathcal{A}_{t_1}(3)| + |\mathcal{A}_{t_2}(3)| = 34 \ll |\mathcal{B}_{t_1}(3)| + |\mathcal{B}_{t_2}(3)| = 592$$

⁴Here, the case of $\beta_{t_1} = \beta_{t_2} = 5$ only has numerical meaning because in this case the rate $\omega = 0$ is trivial.

$$\begin{aligned}
 &\ll 2 \cdot \binom{|E|}{3} = 2 \cdot \binom{21}{3} = 2660, \\
 |\mathcal{A}_{t_1}(4)| + |\mathcal{A}_{t_2}(4)| &= 14 \ll |\mathcal{B}_{t_1}(4)| + |\mathcal{B}_{t_2}(4)| = 858 \\
 &\ll 2 \cdot \binom{|E|}{4} = 2 \cdot \binom{21}{4} = 11916, \\
 |\mathcal{A}_{t_1}(5)| + |\mathcal{A}_{t_2}(5)| &= 2 \ll |\mathcal{B}_{t_1}(5)| + |\mathcal{B}_{t_2}(5)| = 484 \\
 &\ll 2 \cdot \binom{|E|}{5} = 2 \cdot \binom{21}{5} = 40698.
 \end{aligned}$$

C. Quantification of the Improvement

In this subsection, we further quantify the improvement over the existing results by both theoretical analysis and numerical simulations, which show the significance of the improvement. We first prove a lower bound and an upper bound on the obtained upper bound $\sum_{t \in T} |\mathcal{A}_t(\beta_t)|$.

Theorem 18: For a sink node t , let β_t be a nonnegative integer not larger than $C_t - \omega$. Then

$$\binom{|\text{In}(t)|}{\beta_t} \leq |\mathcal{A}_t(\beta_t)| \leq \binom{|\mathcal{A}_t(1)|}{\beta_t}.$$

Proof: By Corollary 15, we obtain the upper bound $\binom{|\mathcal{A}_t(1)|}{\beta_t}$ on $|\mathcal{A}_t(\beta_t)|$. On the other hand, we note that for the sink node t , each size- β_t subset of $\text{In}(t)$ is primary for t . We have thus proved the lower bound $\binom{|\text{In}(t)|}{\beta_t}$ on $|\mathcal{A}_t(\beta_t)|$. The theorem is proved. ■

By Theorem 18, we immediately obtain a corollary of Theorem 16 on the required field size.

Corollary 19: Let $\mathbb{F}_q$ be a finite field of order q . Let T be the set of sink nodes on the network G with $C_t \geq \omega$, $\forall t \in T$, where ω is the rate of the source. For each $t \in T$, let β_t be a nonnegative integer not larger than $C_t - \omega$. Then, there exists an $\mathbb{F}_q$ -valued rate- ω LNEC code on G with the minimum distance at t not smaller than $\beta_t + 1$ for each $t \in T$, i.e., $d_{\min}^{(t)} \geq \beta_t + 1$, $\forall t \in T$, if the field size q satisfies

$$q > \sum_{t \in T} \binom{|\mathcal{A}_t(1)|}{\beta_t}.$$

Continuing from Example 4, the size 17 of $\mathcal{A}_i(\beta_i)$ is lower bounded by $\binom{|\text{In}(t_i)|}{\beta_i} = \binom{5}{2} = 10$ and upper bounded by $\binom{|\mathcal{A}_i(1)|}{\beta_i} = \binom{7}{2} = 21$ for $i = 1, 2$. Even the upper bound $\binom{|\mathcal{A}_i(1)|}{2} = 21$ is also considerably smaller than $|\mathcal{B}_{t_i}(2)| = 99$.

In fact, the improvement of $|\mathcal{A}_t(\beta_t)|$ over $|\mathcal{B}_t(\beta_t)|$ (and also the improvement of $\sum_{t \in T} |\mathcal{A}_t(\beta_t)|$ over $\sum_{t \in T} |\mathcal{B}_t(\beta_t)|$ and $\sum_{t \in T} \binom{|E|}{\beta_t}$) can be unbounded in the difference of the logarithms, or equivalently, the ratio in the exponent. This is illustrated in the following example, where we consider the combination networks which are not only one of the most insightful network models in network coding theory but also of significant interest in commercial disk arrays as a kind of fault-tolerant data storage system [27], [28].

Example 5: Consider the combination network $G_{N,k}$ (see [27, p.26], [28, p.450]), where $k \leq N$. In this network, there is a single source node s , N intermediate nodes, and $|T| = \binom{N}{k}$ sink nodes. Each intermediate node is connected to s , and each sink

node is connected to a distinct subset of k intermediate nodes. The number of the edges in the network is $|E| = N + |T| \cdot k$ and the minimum cut capacity C_t separating every sink node $t \in T$ from s is k , i.e., $C_t = k$, $\forall t \in T$. Further, we take $\beta_t = k/2$, $\forall t \in T$,⁵ i.e., for each sink node, half of the capacity is used as redundancy for error correction.

We now consider the three upper bounds on the minimum required field size for the existence of a rate- ω LNEC code with minimum distance $d_{\min}^{(t)} \geq \beta_t + 1 = k/2 + 1$ for each $t \in T$, where the rate $\omega \leq C_t - \beta_t = k/2$. By a simple calculation, we obtain $|\mathcal{A}_t(\beta_t)|$, $|\mathcal{B}_t(\beta_t)|$ and $\binom{|E|}{\beta_t}$ as follows:

$$|\mathcal{A}_t(\beta_t)| = \binom{k}{\frac{k}{2}}, \quad |\mathcal{B}_t(\beta_t)| = 2^{\frac{k}{2}} \cdot \binom{k}{\frac{k}{2}},$$

and

$$\binom{|E|}{\beta_t} = \binom{N + |T| \cdot k}{\frac{k}{2}} = \binom{N + k \cdot \binom{N}{k}}{\frac{k}{2}}.$$

Then, we readily see that

$$\begin{aligned}
 \log_2 |\mathcal{B}_t(\beta_t)| - \log_2 |\mathcal{A}_t(\beta_t)| &= \log_2 \frac{|\mathcal{B}_t(\beta_t)|}{|\mathcal{A}_t(\beta_t)|} \\
 &= \frac{k}{2} \rightarrow \infty \text{ as } k \rightarrow \infty,
 \end{aligned}$$

which further implies that

$$\log_2 \binom{|E|}{\beta_t} - \log_2 |\mathcal{A}_t(\beta_t)| \rightarrow \infty \text{ as } k \rightarrow \infty.$$

Also, we have randomly generated 60 single-source single-sink networks with 20-80 edges, where the minimum cut capacities separating the sink from the source for the networks are between 4 and 9. Over the networks, we calculate $|\mathcal{A}_t(\beta_t)|$ with the aid of a computer and compare $\log_2 |\mathcal{A}_t(\beta_t)|$ with $\log_2 \binom{|E|}{\beta_t}$ for $1 \leq \beta_t \leq 5$. The numerical results are illustrated in Fig. 6, where we use a pair of plots with the same color and marker to represent $\log_2 \binom{|E|}{\beta_t}$ and $\log_2 |\mathcal{A}_t(\beta_t)|$, respectively, for the same β_t . (Here, the line for $\log_2 \binom{|E|}{\beta_t}$ is always above on the line for $\log_2 |\mathcal{A}_t(\beta_t)|$.) The numerical results show that the improvement of our bound in the difference of the logarithms is in general significant and tends to be larger as β_t increases.

A straightforward consequence of Theorem 16 is an improved upper bound on the minimum required field size for the existence of a rate- ω LNEC MDS code as presented in the following theorem.

Theorem 20: Let $\mathbb{F}_q$ be a finite field of order q and T be the set of sink nodes on the network G with $C_t \geq \omega$, $\forall t \in T$, where ω is the rate of the source. There exists an $\mathbb{F}_q$ -valued rate- ω LNEC MDS code on G if the field size q satisfies

$$q > \sum_{t \in T} |\mathcal{A}_t(\delta_t)|,$$

where $\delta_t \triangleq C_t - \omega$ is called the redundancy of the sink node $t \in T$.

The best known upper bound $\sum_{t \in T} |\mathcal{B}_t(\delta_t)|$ on the minimum required field size for the existence of a rate- ω LNEC MDS code was presented in [10] (cf. [10, Theorem 5]). Similarly,

⁵Without loss of generality, we assume that k is an even number.

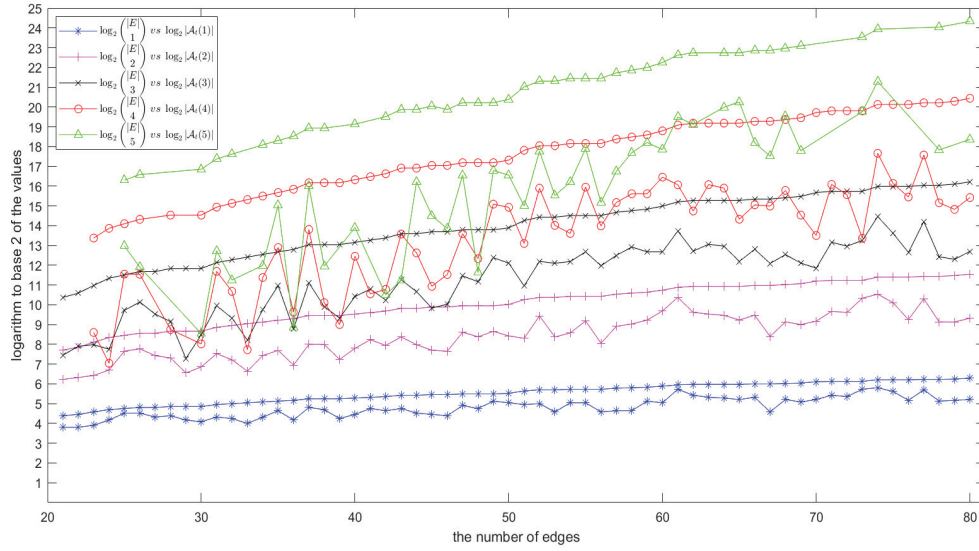


Fig. 6. Numerical comparison.

this best known bound in general is significantly improved by the bound in Theorem 20. In fact, the LNEC code considered in Example 4 is MDS and we have seen that the improvement is significant. Furthermore, similar to Theorem 18, a lower bound and an upper bound on the size of $\mathcal{A}_t(\delta_t)$ is given as follows.

Corollary 21: For a sink node t , the size of $\mathcal{A}_t(\delta_t)$ is lower and upper bounded as follows:

$$\left(\begin{matrix} |\ln(t)| \\ \delta_t \end{matrix} \right) \leq |\mathcal{A}_t(\delta_t)| \leq \left(\begin{matrix} |\mathcal{A}_t(1)| \\ \delta_t \end{matrix} \right).$$

We continue to consider a network as depicted in Fig. 1, where we note that $|\ln(t)| = |E|$ for such a network. By the discussion in the paragraph immediately above Example 4, it is shown that the lower and upper bounds in Corollary 21 are tight for all $\delta_t \leq |\ln(t)|$, i.e.,

$$\left(\begin{matrix} |\ln(t)| \\ \delta_t \end{matrix} \right) = |\mathcal{A}_t(\delta_t)| = \left(\begin{matrix} |\mathcal{A}_t(1)| \\ \delta_t \end{matrix} \right), \quad (27)$$

where $\mathcal{A}_t(1) = \{e : e \in E\}$ and thus $|\mathcal{A}_t(1)| = |\ln(t)| = |E|$. Further, since network error correction coding over such a network depicted in Fig. 1 can be regarded as the model of classical coding theory, $|\mathcal{A}_t(\delta_t)|$, which is equal to $\left(\begin{matrix} |\ln(t)| \\ \delta_t \end{matrix} \right)$ by (27) here, is an upper bound on the minimum required field size for the existence of an $[\ln(t), |\ln(t)| - \delta_t]$ linear MDS code, where $|\ln(t)|$ and $|\ln(t)| - \delta_t$ are the length and dimension of the code, respectively. In general, there exist linear MDS codes over a field of size smaller than this bound. For example, let $|E| = n$ and $\delta_t = n - k$, where k ($k \leq n$) is the designed dimension of the code. There exists an $[n, k]$ linear MDS code over a finite field $\mathbb{F}_q$ if $q \geq n - 1$. Further, there is a longstanding conjecture on the field size for the existence of linear MDS codes, which we present as follows.

MDS Conjecture ([20, Chapter 7.4]): *If there is a nontrivial $[n, k]$ linear MDS code over $\mathbb{F}_q$, then $q \geq n - 1$, except when q is even and $k = 3$ or $k = q - 1$, in which case $q \geq n - 2$.*

Remark 1: For linear codes discussed here, we only consider the network topology (as depicted in Fig. 1) and error correction capability ($d \leq n - k + 1$ with d being the minimum Hamming distance) to bound the required field size for the existence of $[n, k]$ linear codes. In fact, the field size can be further reduced by additionally considering structure of linear codes. However, for LNEC coding discussed in the current paper, we consider the existence of LNEC codes on a network of which any topology is allowed. So it seems impossible to reduce the field size by further considering structure of LNEC codes due to the lack of a code structure applicable to arbitrary network topologies.

D. Efficient Algorithm for Computing the Improved Bound

In Section IV-B, an improved upper bound on the minimum required field size for the existence of LNEC codes has been obtained. The bound thus obtained is graph-theoretic and not in a form which is readily computable. Consequently, the best one can ever hope for is to find an efficient algorithm to compute it. We in this subsection will develop an efficient algorithm to compute this bound.

Let t be a sink node on the network $G = (V, E)$ and r be a nonnegative integer not larger than $C_t - \omega$. We first develop an efficient algorithm for finding all size- r primary edge subsets for t , which compose the collection $\mathcal{A}_t(r)$. An implementation of the algorithm is given in Algorithm 1.

Verification of Algorithm 1:

- 1) In Lines 1 and 2, initialize two sets $\mathcal{A}_t(r)$ and $\mathcal{B}$ to the empty set and the set of all size- r edge subsets of E_t , respectively, where E_t denotes the set of edges in E from which t is reachable, i.e., for each $e \in E_t$, there exists a directed path from e to t on the network G .
- 2) In Lines 4 and 5, arbitrarily choose an edge subset $\eta \in \mathcal{B}$ and find the primary minimum cut separating t from η , denoted by ρ . We note that for each edge subset η , the

Algorithm 1 Algorithm for Computing $\mathcal{A}_t(r)$

Input: The network $G = (V, E)$, a sink node t and a nonnegative integer r .

Output: $\mathcal{A}_t(r)$, the set of all the size- r primary edge subsets for t .

```

begin
1  Set  $\mathcal{A}(r) = \emptyset$ ;
2  Set  $\mathcal{B} = \{\eta \subseteq E_t : |\eta| = r\}$ , where  $E_t$  is the set of
   the edges in  $E$  from which  $t$  is reachable;
   // If there exists a directed path from an edge  $e$  to
    $t$ , we say  $t$  is reachable from  $e$  or  $e$  can reach  $t$ .
3  while  $\mathcal{B} \neq \emptyset$  do
4      choose an edge subset  $\eta$  in  $\mathcal{B}$ ;
5      find the primary minimum cut  $\rho$  separating  $t$ 
       from  $\eta$ ;
       // The primary minimum cut  $\rho$  separating  $t$ 
       from  $\eta$  is a primary edge subset for  $t$ .
6      if  $|\rho| \neq r$  then // Namely,  $|\rho| < r$ .
7          remove  $\eta$  from  $\mathcal{B}$ ;
8      else // Namely,  $|\rho| = r$ .
9          add  $\rho$  to  $\mathcal{A}(r)$ ;
          partition  $E_t$  into two parts  $E_{t,\rho}$  and
           $E_{t,\rho}^c = E_t \setminus E_{t,\rho}$ ;
          // Here,  $E_{t,\rho}$  is the set of the edges from
          which  $t$  is reachable upon deleting the edges
          in  $\rho$ . // Note that  $\rho \subseteq E_{t,\rho}^c$ . for each
           $\mu \in \mathcal{B}$  do
11             if  $\mu \subseteq E_{t,\rho}^c$  then
12                 remove  $\mu$  from  $\mathcal{B}$ ;
13             end
14         end
15     end
16 end
17 Return  $\mathcal{A}(r)$ . // After the “while” loop,  $\mathcal{A}(r)$ 
   contains all the size- $r$  primary edge subsets for  $t$ , i.e.,
    $\mathcal{A}(r) = \mathcal{A}_t(r)$ .
end

```

primary minimum cut separating t from η exists and is unique.

3) According to the fact that

$$|\rho| = \text{mincut}(\eta, t) \leq |\eta| = r, \quad (28)$$

we consider two cases below.

Case 1: If $|\rho| \neq r$, which implies $|\rho| < r$ by (28), then the “if” statement (Line 7) is executed. In this case, we readily see that ρ is not a size- r primary edge subset for t . Then, we remove η from $\mathcal{B}$ and go back to Line 3 for checking whether the updated $\mathcal{B}$ is empty or not.

Case 2: If $|\rho| = r$, which implies that ρ is a size- r primary edge subset for t , then the “else” statement (Lines 8–12) is executed. To be specific, in Line 8, add this size- r primary edge subset ρ to $\mathcal{A}(r)$. In Line 9, partition the edge set E_t into two disjoint subsets: $E_{t,\rho}$ and $E_{t,\rho}^c \triangleq E_t \setminus E_{t,\rho}$, where $E_{t,\rho}$ is the set

Algorithm 2 Algorithm for Partitioning E_t Into $E_{t,\rho}$ and $E_{t,\rho}^c$

Input: The network $G = (V, E)$ and an edge subset $\rho \subseteq E$.

Output: $E_{t,\rho}$, the set of all the edges that can reach t upon deleting the edges in ρ .

```

begin
1  Unmark all nodes in  $V$ ;
2  mark sink node  $t$ ;
3  set an edge-set E-SET =  $\emptyset$ ;
4  set a node-set N-SET =  $\{t\}$ ;
5  while N-SET  $\neq \emptyset$  do
6      select a node  $a$  in N-SET;
7      for each node  $v \in V$  incident to an edge  $(v, a)$ 
       not in  $\rho$  do
8          add all parallel edges leading from  $v$  to  $a$ 
       and not in  $\rho$  to E-SET;
9          if  $v$  is unmarked then
10             mark node  $v$ ;
11             add node  $v$  to N-SET;
12         end
13     end
14 delete node  $a$  from N-SET;
15 end
16 Return E-SET. // After the “while” loop, E-SET
   contains all the edges that can reach  $t$  upon deleting
   the edges in  $\rho$ , i.e., E-SET =  $E_{t,\rho}$ .
end

```

of edges from which t is reachable upon deleting the edges in ρ . Note that $\rho \subseteq E_{t,\rho}^c$. Next, for the “for” loop (Lines 10–12), all the edge subsets in $\mathcal{B}$ that are subsets of $E_{t,\rho}^c$ are removed. By definition, each edge subset η in $\mathcal{B}$, regardless of whether $\text{mincut}(\eta, t) = r$ or $\text{mincut}(\eta, t) < r$, is a subset of $E_{t,\rho}^c$ if and only if ρ separates t from η (cf. [9, Lemma A.1]). This immediately implies that after this “for” loop, all the edge subsets in $\mathcal{B}$ from which ρ separates t are removed from $\mathcal{B}$, and none of the other size- r primary edge subsets are removed from $\mathcal{B}$. Thus, we see that in each iteration, exactly one size- r primary edge subset for t is added to $\mathcal{A}(r)$.

4) Repeat Steps 2) and 3) above until $\mathcal{B}$ is empty and output $\mathcal{A}(r)$ in Line 13, which is now equal to $\mathcal{A}_t(r)$.

Remark 2: In fact, in Algorithm 1, it is sufficient to initialize

$$\mathcal{B} = \{\eta \subseteq E_t : |\eta| = \text{mincut}(\eta, t) = r\}.$$

As such, for each $\eta \in \mathcal{B}$, the primary minimum cut separating t from η is a size- r primary edge subset for t , and then we do not need “if” statement in Line 6 any more. However, for this case, it is necessary to determine $\mathcal{B}$ in advance, namely that find out all size- r regular- t edge subsets. This would cost high complexity. Thus, for the sake of computational efficiency, here we only initialize $\mathcal{B} = \{\eta \subseteq E_t : |\eta| = r\}$ and add the “if” statement in Line 6 to judge whether the primary minimum cut ρ found in Line 5 is of size r or not.

In Algorithm 1, the two crucial steps are i) to find the primary minimum cut ρ separating t from an edge subset η in $\mathcal{B}$ (Line 5), and ii) to partition E_t into $E_{t,\rho}$ and $E_{t,\rho}^c$ (Line 9). The step i) can be efficiently implemented by slightly modifying the foregoing Algorithm 4 for finding the top of the lattice ($\text{MinCut}(s, t), \leq$). To be specific, on the network G , we first subdivide each edge $e \in \eta$ by creating a node v_e for e and splitting e into two edges $e^{(1)}$ and $e^{(2)}$ with $\text{tail}(e^{(1)}) = \text{tail}(e)$, $\text{head}(e^{(2)}) = \text{head}(e)$, and $\text{head}(e^{(1)}) = \text{tail}(e^{(2)}) = v_e$. Then, we create a new node v_η and add a new edge from v_η to v_e for every node v_e , $e \in \eta$. By definition, we can readily see that a cut separating t from v_η is a cut separating t from η , and vice versa (where, whenever $e^{(1)}$ or $e^{(2)}$ appears in the cut, replace it by e). As such, for the purpose of finding the primary minimum cut separating t from η on G , we only need to consider algorithms for finding the top of the lattice ($\text{MinCut}(v_\eta, t), \leq$), i.e., the minimum cut separating t from v_η that is “nearest” to t .

Next, we consider the step ii), i.e., the step of partitioning E_t into $E_{t,\rho}$ and $E_{t,\rho}^c$. Toward this end, it suffices to determine the edge set $E_{t,\rho}$, i.e., find out all the edges that can reach t upon deleting the edges in ρ . This can be implemented efficiently by the following Algorithm 2.

Verification of Algorithm 2:

- 1) Algorithm 2 extends from the sink node t and identifies an increasing number of edges that can reach t . At any point during the execution of the algorithm, all the nodes in the network can be in one of two states: marked or unmarked. The marked nodes are those from which t is reachable upon deleting the edges in ρ , and the unmarked nodes are those yet to be classified. The edges in the set E-SET at this point have been identified to be those from which t is reachable. The set N-SET contains marked nodes whose input edges have not been processed.
- 2) When a node $a \in \text{N-SET}$ is selected in Line 6, all the input edges of a that are not in ρ are added to E-SET in the “for” loop (Lines 7–11). Since $a \in \text{N-SET}$, we see that a is marked and so t is reachable from a . This implies that t is reachable from all these input edges and they are added to E-SET in Line 8. The node v incident to an edge (v, a) can reach t via node a . If v is unmarked, then mark v in Line 10. Otherwise, v has already been marked and so t is reachable from v . After the “for” loop (Lines 7–11), all the input edges of a that are not in ρ are added to E-SET and all the nodes v incident to an edge (v, a) are marked. Now, the node a has been processed and is removed from N-SET in Line 12. The algorithm terminates when the set of nodes N-SET is empty. At this point, all the nodes that can reach t have been marked and processed, and the edge set E-SET contains all the edges that can reach t upon deleting the edges in ρ , namely that $\text{E-SET} = E_{t,\rho}$.

Next, we consider the complexity of Algorithm 2. We can readily see that the algorithm traverses all the edges in $E_{t,\rho}$ exactly once, and thus Algorithm 2 can find the edge set $E_{t,\rho}$ in $\mathcal{O}(|E_{t,\rho}|)$ time.

Finally, by combining the foregoing discussions, we analyze the total complexity of Algorithm 1 for computing $\mathcal{A}_t(r)$ as follows: i) The complexity for finding the primary minimum cut ρ separating t from an edge subset η (Line 5 in Algorithm 1) is at most $\mathcal{O}(|E_t|)$; ii) The complexity for partitioning E_t into two parts $E_{t,\rho}$ and $E_{t,\rho}^c$ (Line 9 in Algorithm 1) is at most $\mathcal{O}(|E_{t,\rho}|)$, not larger than $\mathcal{O}(|E_t|)$; iii) Removing all the edge subsets in $\mathcal{B}$ that are subsets of $E_{t,\rho}^c$ (Lines 10–12 in Algorithm 1) can be implemented by creating an appropriate data structure to avoid computational complexity; iv) The “while” loop (Line 3 in Algorithm 1) is executed $|\mathcal{A}_t(r)|$ times.⁶ So the complexity of Algorithm 1 is at most $\mathcal{O}(|\mathcal{A}_t(r)| \cdot |E_t|)$, that is linear time in $|E_t|$.

Remark 3: We note that the previous bound $\sum_{t \in T} |\mathcal{B}_t(\beta_t)|$ (see Proposition 4), which is also graph-theoretic, is not in explicit form. So, in order to compute this bound, we also need to find an efficient algorithm. Toward this end, we need to compute the minimum cut capacity for each of the size- β_t edge subsets of E_t . Under the consideration of the standard path augmentation algorithm, the complexity for computing the minimum cut capacity of a size- β_t edge subset is at most $\mathcal{O}(\beta_t \cdot |E_t|)$, or $\mathcal{O}(|E_t|)$ if we regard β_t as a constant. Then, the total complexity for computing $|\mathcal{B}_t(\beta_t)|$ is $\mathcal{O}\left(\binom{|E_t|}{\beta_t} \cdot |E_t|\right)$, which is larger than $\mathcal{O}(|\mathcal{A}_t(\beta_t)| \cdot |E_t|)$, the complexity of Algorithm 1 for computing $|\mathcal{A}_t(\beta_t)|$. This hence shows that the improvement of our bound $\sum_{t \in T} |\mathcal{A}_t(\beta_t)|$ over the existing bound $\sum_{t \in T} |\mathcal{B}_t(\beta_t)|$ is not only in the value but also in computational complexity.

V. CONCLUSION

In this paper, we have investigated the underlying mathematical structure of edge subsets on a finite directed acyclic graph using an order-theoretic approach. We proved that a collection of edge subsets, depending on different conditions it satisfies, together with the corresponding “cut-separating” partial order, can form a meet-semilattice, a join-semilattice or a lattice. The bottom and/or top elements thus induced generalize the concept of the primary minimum cut studied in the literature. We provided a new lattice-theoretic perspective to understand the primary minimum cut and prove its existence and uniqueness. By slightly modifying the algorithm to find primary minimum cut (see Algorithm 6 in [22]), we further presented efficient algorithms to find the top and/or the bottom, of which the computational complexity is in a linear time of the number of edges. The introduced concepts and obtained results as a bridge connect graph theory and lattice theory, which appear to be of fundamental interest in graph theory, lattice theory, and even beyond.

By applying the approach of the developed edge-subset (semi-)lattice, we proved an improved upper bound on the minimum required field size for the existence of LNEC (MDS) codes. In LNEC coding, the minimum required field size for the existence of LNEC (MDS) codes is an open problem not only of theoretical interest but also of practical

⁶Here, it suffices to consider edge subsets $\eta \in \mathcal{B}$ with $\text{mincut}(\eta, t) = r$.

TABLE I
 NOTATIONS FREQUENTLY USED IN THE PAPER

Notation	Explanation
$G = (V, E)$	A finite directed acyclic graph, where V and E are the sets of nodes and edges, respectively.
s	The single source node.
t	A sink node.
T	The set of sink nodes.
r	A positive integer.
$\text{head}(e)$	The head of an edge e .
$\text{tail}(e)$	The tail of an edge e .
$\text{Out}(v)$	The set of output edges of a node v .
$\text{In}(v)$	The set of input edges of a node v .
ρ, ξ, η	Edge subsets.
$G^\top$	The reverse graph of G .
$\vee$	The binary relation of join, see the paragraph above Definition 2.
$\wedge$	The binary relation of meet, see the paragraph above Definition 2.
$\perp$	The bottom of a partially ordered set, see Definition 3.
$\top$	The top of a partially ordered set, see Definition 3.
$\text{mincut}(s, \rho)$	The minimum cut capacity separating ρ from s , see the second paragraph in Section III-B.
$\text{mincut}(\rho, t)$	The minimum cut capacity separating t from ρ , see the paragraph above Theorem 8.
$\text{mincut}(s, t)$	The minimum cut capacity separating t from s , see the second paragraph in Appendix B.
$\text{MinCut}(s, t)$	The set of all the minimum cuts separating t from s , see Lemma 5.
s -regular	An edge subset ρ is s -regular if $ \rho = \text{mincut}(s, \rho)$.
regular- t	An edge subset ρ is regular- t if $ \rho = \text{mincut}(\rho, t)$.
$s \sim$	For two edge subsets ρ and η , $\rho \sim \eta$ if and only if ρ and η have a common minimum cut ξ separating each of them from s . See the second paragraph in Section III-B.
$\sim_t$	For two edge subsets ρ and η , $\rho \sim_t \eta$ if and only if ρ and η have a common minimum cut ξ separating t from each of them. See the paragraph above Theorem 8.
$\mathcal{B}_s(r)$	The collection of all the s -regular edge subsets of size r , see (1).
$\mathcal{B}_t(r)$	The collection of all the regular- t edge subsets of size r , see (10).
Cl_r	An equivalence class of $\mathcal{B}_s(r)$ under the equivalence relation “ $s \sim$ ”.
cl_r	An equivalence class of $\mathcal{B}_t(r)$ under the equivalence relation “ $\sim_t$ ”.
$\mathcal{A}_t(r)$	The set of the top elements of the join-semilattices $(\text{cl}_r, \leq_t)$ for all the equivalence classes cl_r of $\mathcal{B}_t(r)$, see (11) or (12).
$s \leq$	For $\rho, \eta \in \mathcal{B}_s(r)$, $\rho \leq \eta$ if and only if no path exists from s to any edge in η upon deleting the edges in ρ . See the paragraph above Example 2.
$\leq_t$	For $\rho, \eta \in \mathcal{B}_t(r)$, $\rho \leq_t \eta$ if and only if no path exists from any edge in ρ to t upon deleting the edges in η . See the paragraph above Theorem 8.
ω	The information rate of the source in LNEC coding.
$\Phi(t)$	The received codebook at t , see (14).
$\Delta(t, \rho)$	The error space of an edge subset ρ at a sink node t , see (15).
$d_{\min}^{(t)}$	The minimum distance of the received codebook $\Phi(t)$ at t , see (16).
C_t	The minimum cut capacity separating t from s , see (17).
$\mathcal{H}(r)$	The collection of all the edge subsets of size up to r , see (19).
$G_{N,k}$	A combination network with a single source node s , N intermediate nodes and each sink node is connected to a distinct subset of k intermediate nodes. See Example 5.

importance. We further quantified the improvement over the existing results by both theoretical analysis and numerical simulations. This shows that the improvement is in general significant. The improved upper bound depends only on the network topology and the required error correction capability but not on a specific code construction. Based on the developed algorithms for finding the top and/or the bottom of the (semi-)lattice, we also developed an efficient algorithm that can compute the bound in a linear time of the number of edges.

APPENDIX

A. Notations

See Table I.

B. Algorithms for Finding the Bottom and Top

We continue to consider the graph $G = (V, E)$ with the source node s and the sink node t (cf. the first paragraph in Section III-B). Here, we only focus on developing approach to efficiently find the top and the bottom of the edge-subset lattice $(\text{MinCut}(s, t), \leq)$, and then we can slightly modify the

developed approach to efficiently find the top of an edge-subset join-semilattice $(\text{cl}, \leq_t)$ and the bottom of an edge-subset meet-semilattice $(\text{Cl}, \leq_s)$.

Recall that $\text{mincut}(s, t)$ denotes the minimum cut capacity separating t from s on G . By the max-flow min-cut theorem [1], [2], the value $v(F)$ of a maximum flow F from s to t is equal to the minimum cut capacity $\text{mincut}(s, t)$, i.e., $v(F) = \text{mincut}(s, t)$. Since all the edges on the graph G is assumed to have unit-capacity, $\text{mincut}(s, t)$ is a positive integer and the maximum flow F can be decomposed into $\text{mincut}(s, t)$ edge-disjoint paths from s to t . Such $\text{mincut}(s, t)$ edge-disjoint paths can be found in polynomial time in $|E|$ (cf. [29], [30]). We in the following present Algorithms 3 and 4, which are used to find the bottom $\perp$ and the top $\top$ of the lattice $(\text{MinCut}(s, t), \leq)$, respectively.

Next, we give an example to illustrate Algorithm 3 and 4.

Example 6: Consider a directed acyclic graph G with a maximum flow F from the source node s to the sink node t as illustrated in Fig. 7. First, we illustrate Algorithm 3 in Fig. 8 that outputs the bottom $\perp$ of the lattice $(\text{MinCut}(s, t), \leq)$.

Algorithm 3 Algorithm for Finding the Bottom $\perp$ of the Lattice ($\text{MinCut}(s, t), \leq$)

Input: The graph $G = (V, E)$ with a maximal flow F from the source node s to the sink node t . For every edge e in the corresponding mincut(s, t) edge-disjoint paths, the flow value is equal to 1, i.e., $F(e) = 1$; otherwise, the flow value is equal to 0, i.e., $F(e) = 0$.

Output: The bottom $\perp$ of ($\text{MinCut}(s, t), \leq$).

```

begin
1  Set  $U_{\perp} = \{s\}$ ;
2  for each newly-added node  $u \in U_{\perp}$  do
3      for each node  $v \in V \setminus U_{\perp}$  do
4          if either  $\exists$  a (forward) edge  $e$  from  $u$  to  $v$ 
             s.t.  $F(e) = 0$  or  $\exists$  a (reverse) edge  $e$  from  $v$ 
             to  $u$  s.t.  $F(e) = 1$  then
5              replace  $U_{\perp}$  by  $U_{\perp} \cup \{v\}$ .
             end
          end
      end
  end
  // The "for" loop (Lines 2-5) terminates when
  no more new node is added to  $U_{\perp}$  and all the nodes
  in  $U_{\perp}$  have been processed. Return
   $\perp = \{e : \text{tail}(e) \in U_{\perp} \text{ and } \text{head}(e) \in V \setminus U_{\perp}\}$ .
end

```

Algorithm 4 Algorithm for Finding the Top $\top$ of the Lattice ($\text{MinCut}(s, t), \leq$)

Input: The graph $G = (V, E)$ with a maximal flow F from the source node s to the sink node t . For every edge e in the corresponding mincut(s, t) edge-disjoint paths, the flow value is equal to 1, i.e., $F(e) = 1$; otherwise, the flow value is equal to 0, i.e., $F(e) = 0$.

Output: The top $\top$ of ($\text{MinCut}(s, t), \leq$).

```

begin
1  Set  $U_{\top} = \{t\}$ ;
2  for each newly-added node  $u \in U_{\top}$  do
3      for each node  $v \in V \setminus U_{\top}$  do
4          if either  $\exists$  a (reverse) edge  $e$  from  $v$  to  $u$  s.t.
              $F(e) = 0$  or  $\exists$  a (forward) edge  $e$  from  $u$  to
              $v$  s.t.  $F(e) = 1$  then
5              replace  $U_{\top}$  by  $U_{\top} \cup \{v\}$ .
             end
          end
      end
  end
  // The "for" loop (Lines 2-5) terminates when
  no more new node is added to  $U_{\top}$  and all the nodes
  in  $U_{\top}$  have been processed. Return
   $\top = \{e : \text{tail}(e) \in V \setminus U_{\top} \text{ and } \text{head}(e) \in U_{\top}\}$ .
end

```

- Algorithm 3 starts with $U_{\perp} = \{s\}$. First, for $s \in U_{\perp}$ regarded as the initial newly-added node, we see that only the edge (s, i_3) satisfies the condition of the "if" statement in Line 4 of Algorithm 3, i.e., (s, i_3) is a (forward) edge

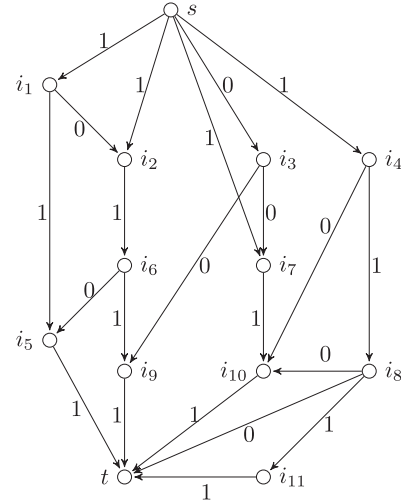


Fig. 7. The directed acyclic graph G .

from s to i_3 with flow value 0. Then, after the "for" loop in Lines 3-5, $U_{\perp}$ is updated to $\{s, i_3\}$. Next, for the newly-added node $i_3 \in U_{\perp}$, only the edges (i_3, i_7) and (i_3, i_9) satisfy the condition of the "if" statement in Line 4. To be specific, (i_3, i_7) and (i_3, i_9) are two (forward) edges with flow values 0 from i_3 to i_7 and i_9 , respectively. After the "for" loop (Lines 3-5), $U_{\perp}$ is updated to $\{s, i_3, i_7, i_9\}$. This is illustrated in Fig. 8(a).

- For the newly-added node $i_7 \in U_{\perp}$, the edge (i_7, i_{10}) is the only edge incident to i_7 with another node in $V \setminus U_{\perp}$. But (i_7, i_{10}) is a (forward) edge from i_7 to i_{10} with flow value 1. This does not satisfy the condition of the "if" statement in Line 4. For the newly-added node $i_9 \in U_{\perp}$, (i_9, t) and (i_6, i_9) are all the edges incident to i_9 . The edge (i_9, t) is a (forward) edge but its flow value is 1; and (i_6, i_9) is a (reverse) edge with flow value 1. Thus, the condition of the "if" statement in Line 4 of Algorithm 3 is satisfied only for the edge (i_6, i_9) . Update $U_{\perp}$ to $\{s, i_3, i_6, i_7, i_9\}$. This is illustrated in Fig. 8(b).
- For the newly-added node $i_6 \in U_{\perp}$, after the "for" loop in Lines 3-5, $U_{\perp}$ is already updated to $\{s, i_2, i_3, i_5, i_6, i_7, i_9\}$. This is illustrated in Fig. 8(c).
- So on and so forth, we obtain

$$U_{\perp} = \{s, i_1, i_2, i_3, i_5, i_6, i_7, i_9\}$$

after processing the newly-added nodes i_2 and i_5 , as illustrated in Fig. 8(d). Next, for the newly-added node i_1 , no edge incident to i_1 with another node in $V \setminus U_{\perp}$ satisfies the condition of the "if" statement in Line 4. Now, no more node is added to $U_{\perp}$ and all the nodes in $U_{\perp}$ have been processed. The "for" loop (Lines 2-5) terminates. Return the edge set

$$\begin{aligned} \perp &= \{e : \text{tail}(e) \in U_{\perp} \text{ and } \text{head}(e) \in V \setminus U_{\perp}\} \\ &= \{(i_5, t), (i_9, t), (i_7, i_{10}), (s, i_4)\}, \end{aligned}$$

and Algorithm 3 terminates. We readily check that $\perp$ is the bottom of $(\text{MinCut}(s, t), \leq)$, namely the minimum cut separating t from s that is "nearest" to s .

Next, we illustrate Algorithm 4 in Fig. 9 that outputs the top $\top$ of the lattice $(\text{MinCut}(s, t), \leq)$.

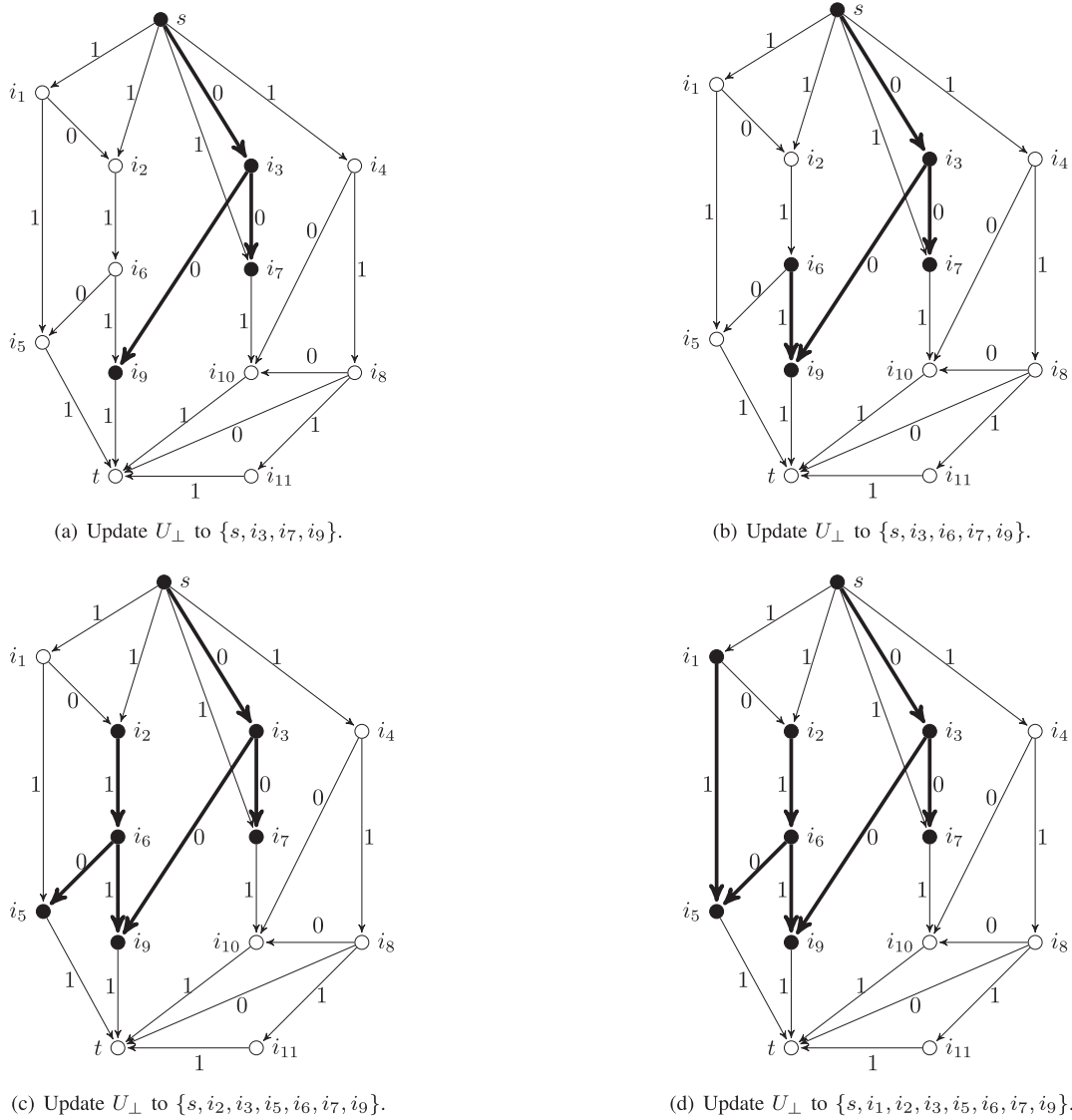


Fig. 8. An example to use Algorithm 3 for finding the bottom $\perp$ of $(\text{MinCut}(s, t), \leq)$.

- Algorithm 4 starts with $U_{\top} = \{t\}$. First, for the newly-added node $t \in U_{\top}$, only the edge (i_8, t) satisfies the condition of the “if” statement in Line 4 of Algorithm 4, i.e., (i_8, t) is a (reverse) edge from i_8 to t with flow value 0. After the “for” loop in Lines 3-5, $U_{\top}$ is updated to $\{t, i_8\}$. This is illustrated in Fig. 9(a).
- Next, for the newly-added node $i_8 \in U_{\top}$, (i_4, i_8) , (i_8, i_{10}) , (i_8, i_{11}) are all the edges incident to i_8 with another node in $V \setminus U_{\top}$. The edge (i_4, i_8) is a (reverse) edge with flow value 1; (i_8, i_{10}) is a (forward) edge with flow value 0; and (i_8, i_{11}) is a (forward) edge with flow value 1. So the condition of the “if” statement in Line 4 of Algorithm 4 is satisfied only for the edge (i_8, i_{11}) . Consequently, after the “for” loop (Lines 3-5), $U_{\top}$ is updated to $\{t, i_8, i_{11}\}$. This is illustrated in Fig. 9(b). After that, for the newly-added node $i_{11} \in U_{\top}$, there is no more edges satisfies the condition of the “if” statement in Line 4 of Algorithm 4. Now, no more new node is added to $U_{\top}$ and all the nodes in $U_{\top} = \{t, i_8, i_{11}\}$ have been already processed.

Then, return the edge set

$$\begin{aligned} \top &= \{e : \text{tail}(e) \in V \setminus U_{\top} \text{ and } \text{head}(e) \in U_{\top}\} \\ &= \{(i_5, t), (i_9, t), (i_{10}, t), (i_4, i_8)\}, \end{aligned}$$

and Algorithm 4 terminates. We can readily check that $\top$ is the top of $(\text{MinCut}(s, t), \leq)$, namely the minimum cut separating t from s that is “nearest” to the sink node t .

Remark 4: In fact, the bottom (resp. the top) of the lattice $(\text{MinCut}(s, t), \leq)$ on G is the top (resp. the bottom) of the lattice $(\text{MinCut}(t, s), \leq)$ on the reverse graph $G^{\top}$, where we use $\text{MinCut}(t, s)$ to stand for the set of all the minimum cuts separating s from t on $G^{\top}$. So we can execute Algorithm 3 on the reverse graph $G^{\top}$ to find the top of the lattice $(\text{MinCut}(s, t), \leq)$ on G . Here, by a minor modification of Algorithm 3, we obtain Algorithm 4 which directly finds the top of the lattice $(\text{MinCut}(s, t), \leq)$ to avoid reversing the graph G to $G^{\top}$ and then executing Algorithm 3.

The verification of Algorithm 3 (also Algorithm 4) has been given by Guang and Yeung in [22]. We refer the reader to

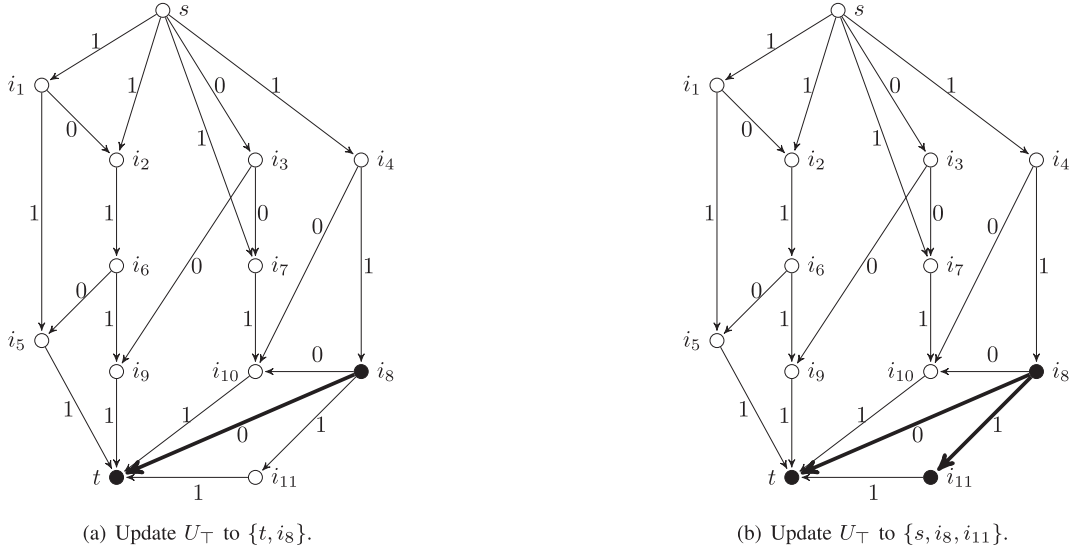


Fig. 9. An example to use Algorithm 4 for finding the top T of $(\text{MinCut}(s, t), \leq)$.

Appendix D in [22] for details. More interestingly, Algorithm 3 can be viewed as the last step of the path augmentation approach [1], [2] (see also [29, Chapter 6.5] and [30, Chapter 7.2]) for finding the maximum flow from a node s to another one t on a directed graph. The last step determines the termination of the path augmentation approach. In other words, after the execution of a path augmentation algorithm, e.g., the Ford-Fulkerson Algorithm [2], the Edmonds-Karp Algorithm [31] and the Dinic Algorithm [32] (also refer to the two comprehensive books [29] and [30] for the algorithms), we are also able to obtain the bottom of the lattice $(\text{MinCut}(s, t), \leq)$, namely the minimum cut separating t from s that is “nearest” to s .

The computational complexity of Algorithm 3 (also Algorithm 4) is at most $\mathcal{O}(|E|)$ since in the algorithm, each edge in E is examined at most once. Furthermore, if we use the path augmentation algorithm to find mincut(s, t) edge-disjoint paths from s to t , then Algorithm 3 is already incorporated as discussed above. Thus, the total complexity for finding the bottom of the lattice $(\text{MinCut}(s, t), \leq)$ is at most $\mathcal{O}(\text{mincut}(s, t) \cdot |E|)$, because one path augmentation requires at most $\mathcal{O}(|E|)$ time as mentioned and the number of the path augmentations is upper bounded by the minimum cut capacity $\text{mincut}(s, t)$.⁷

REFERENCES

- [1] P. Elias, A. Feinstein, and C. Shannon, “A note on the maximum flow through a network,” *IEEE Trans. Inf. Theory*, vol. IT-2, no. 4, pp. 117–119, Dec. 1956.
- [2] L. R. Ford and D. R. Fulkerson, “Maximal flow through a network,” *Can. J. Math.*, vol. 8, no. 3, pp. 399–404, 1956.
- [3] R. Ahlswede, N. Cai, S.-Y. R. Li, and R. W. Yeung, “Network information flow,” *IEEE Trans. Inf. Theory*, vol. 46, no. 4, pp. 1204–1216, Jul. 2000.
- [4] N. Cai and R. W. Yeung, “Network coding and error correction,” in *Proc. IEEE Inf. Theory Workshop*, Oct. 2002, pp. 119–122.
- [5] R. W. Yeung and N. Cai, “Network error correction, part I: Basic concepts and upper bounds,” *Commun. Inf. Syst.*, vol. 6, no. 1, pp. 19–36, 2006.
- [6] N. Cai and R. W. Yeung, “Network error correction, II: Lower bounds,” *Commun. Inf. Syst.*, vol. 6, no. 1, pp. 37–54, 2006.
- [7] Z. Zhang, “Linear network error correction codes in packet networks,” *IEEE Trans. Inf. Theory*, vol. 54, no. 1, pp. 209–218, Jan. 2008.
- [8] S. Yang, R. W. Yeung, and Z. Zhang, “Weight properties of network codes,” *Eur. Trans. Telecommun.*, vol. 19, no. 4, pp. 371–383, 2008.
- [9] X. Guang and R. W. Yeung, “A revisit of linear network error correction coding,” *IEEE J. Sel. Areas Inf. Theory*, vol. 4, pp. 514–523, 2023.
- [10] X. Guang, F.-W. Fu, and Z. Zhang, “Construction of network error correction codes in packet networks,” *IEEE Trans. Inf. Theory*, vol. 59, no. 2, pp. 1030–1047, Feb. 2013.
- [11] S. Yang, R. W. Yeung, and C. K. Ngai, “Refined coding bounds and code constructions for coherent network error correction,” *IEEE Trans. Inf. Theory*, vol. 57, no. 3, pp. 1409–1424, Mar. 2011.
- [12] X. Guang and Z. Zhang, *Linear Network Error Correction Coding*. New York, NY, USA: Springer, 2014.
- [13] R. Matsumoto, “Construction algorithm for network error-correcting codes attaining the Singleton bound,” *IEICE Trans. Fundamentals Electron., Commun. Comput. Sci.*, vols. E90–A, no. 9, pp. 1729–1735, Sep. 2007.
- [14] X. Guang, F.-W. Fu, and Z. Zhang, “Variable-rate linear network error correction MDS codes,” *IEEE Trans. Inf. Theory*, vol. 62, no. 6, pp. 3147–3164, Jun. 2016.
- [15] Z. Zhang, “Theory and applications of network error correction coding,” *Proc. IEEE*, vol. 99, no. 3, pp. 406–420, Mar. 2011.
- [16] H. Balli, X. Yan, and Z. Zhang, “On randomized linear network codes and their error correction capabilities,” *IEEE Trans. Inf. Theory*, vol. 55, no. 7, pp. 3148–3160, Jul. 2009.
- [17] R. Koetter and F. R. Kschischang, “Coding for errors and erasures in random network coding,” *IEEE Trans. Inf. Theory*, vol. 54, no. 8, pp. 3579–3591, Aug. 2008.
- [18] D. Silva, F. R. Kschischang, and R. Kötter, “A rank-metric approach to error control in random network coding,” *IEEE Trans. Inf. Theory*, vol. 54, no. 9, pp. 3951–3967, Sep. 2008.
- [19] F. J. MacWilliams and N. J. A. Sloane, *The Theory of Error Correcting Codes*. Amsterdam, The Netherlands: North-Holland, 1977.
- [20] W. C. Huffman and V. Pless, *Fundamentals of Error Correcting Codes*. Cambridge, U.K.: Cambridge Univ. Press, 2003.
- [21] C. E. Shannon, “A mathematical theory of communication,” *Bell Syst. Tech. J.*, vol. 27, no. 3, pp. 379–423, 1948.
- [22] X. Guang and R. W. Yeung, “Alphabet size reduction for secure network coding: A graph theoretic approach,” *IEEE Trans. Inf. Theory*, vol. 64, no. 6, pp. 4513–4529, Jun. 2018.
- [23] X. Guang, R. W. Yeung, and F.-W. Fu, “Local-encoding-preserving secure network coding,” *IEEE Trans. Inf. Theory*, vol. 66, no. 10, pp. 5965–5994, Oct. 2020.
- [24] B. A. Davey and H. A. Priestley, *Introduction to Lattices and Order*, 2nd ed., Cambridge, U.K.: Cambridge Univ. Press, 1990.

⁷Here, we assume that the complexity of each examination of an edge is $\mathcal{O}(1)$.

- [25] G. Birkhoff, *Lattice Theory*, vol. 25, 3rd ed., Providence, RI, USA: AMS Colloquium Publications, 1967.
- [26] X. Guang, J. Lu, and F.-W. Fu, "Small field size for secure network coding," *IEEE Commun. Lett.*, vol. 19, no. 3, pp. 375–378, Mar. 2015.
- [27] R. W. Yeung, S.-Y. R. Li, N. Cai, and Z. Zhang, "Network coding theory," *Found. Trends Commun. Inf. Theory*, vol. 2, nos. 4–5, pp. 241–381, 2005.
- [28] R. W. Yeung, *Information Theory and Network Coding*. New York, NY, USA: Springer, 2008.
- [29] R. K. Ahuja, T. L. Magnanti, and J. B. Orlin, *Network Flows: Theory, Algorithms, and Applications*. Englewood Cliffs, NJ, USA: Prentice-Hall, 1993.
- [30] J. A. Bondy and U. S. R. Murty, *Graph Theory*. Cham, Switzerland: Springer, 2008.
- [31] J. Edmonds and R. M. Karp, "Theoretical improvements in algorithmic efficiency for network flow problems," *J. ACM*, vol. 19, no. 2, pp. 248–264, Apr. 1972.
- [32] E. A. Dinic, "Algorithm for solution of a problem of maximum flow in networks with power estimation," *Sov. Math. Doklady*, vol. 11, pp. 1277–1280, Jan. 1970.

Xuan Guang (Member, IEEE) received the Ph.D. degree in mathematics from the Chern Institute of Mathematics, Nankai University, in 2012. He was a Joint Ph.D. Student with the Ming Hsieh Department of Electrical Engineering, University of Southern California, from January 2011 to August 2012. From November 2015 to November 2018, he was a Post-Doctoral Fellow with the Institute of Network Coding (INC), The Chinese University of Hong Kong, under Hong Kong Scholars Program.

He is currently a Full Professor with the School of Mathematical Sciences and the Key Laboratory of Pure Mathematics and Combinatorics (LPMC), Nankai University. His research interests include information theory and coding theory for function computation, network coding, and information-theoretic security. He was selected into the Hundred Young Academic Leaders Program of Nankai University in 2017. He was a recipient of the 2018 Chinese Information Theory Young Rising Star Award by China Information Theory Society.

Ruopu Cui received the B.S. degree in mathematics from Dalian University of Technology, Dalian, China, in 2020. He is currently pursuing the Ph.D. degree with the School of Mathematical Sciences and LPMC, Nankai University. His research interests include information theory, network coding, network function computation, and network error correction coding.

Raymond W. Yeung (Fellow, IEEE) was born in Hong Kong in June 1962. He received the B.S., M.Eng., and Ph.D. degrees in electrical engineering from Cornell University, Ithaca, NY, USA, in 1984, 1985, and 1988, respectively.

He was on leave with the École Nationale Supérieure des Télécommunications, Paris, France, in Fall 1986. He was a Member of the Technical Staff with AT&T Bell Laboratories from 1988 to 1991. He has held visiting positions with Cornell University, Nankai University, Bielefeld University, the University of Copenhagen, Tokyo Institute of Technology, Munich University of Technology, and Columbia University. He was a Consultant in a project of the Jet Propulsion Laboratory, Pasadena, CA, USA, for salvaging the malfunctioning Galileo Spacecraft, and a Consultant with NEC, USA. Since 1991, he has been with The Chinese University of Hong Kong, where he is currently a Choh-Ming Li Professor of information engineering and the Co-Director of the Institute of Network Coding. His 25-bit synchronization marker was used onboard the Galileo Spacecraft for image synchronization. He is the author of the textbooks *A First Course in Information Theory* (Kluwer Academic/Plenum 2002) and its revision *Information Theory and Network Coding* (Springer 2008), which have been adopted by over 100 institutions around the world. This book has also been published in Chinese (Higher Education Press 2011, translation by Ning Cai et al.) He also co-authored with Shenghao Yang the monograph *BATS Codes: Theory and Applications* (Morgan & Claypool Publishers, 2017). In Spring 2014, he gave the first MOOC on information theory that reached over 25,000 students. His research interests include information theory and network coding.

Dr. Yeung was a member of the Board of Governors of the IEEE Information Theory Society from 1999 to 2001. He is a fellow of Hong Kong Institution of Engineers, Hong Kong Academy of Engineering, and the U.S. National Academy of Inventors. He was a recipient of the Croucher Foundation Senior Research Fellowship in 2000 and 2001, the 2005 IEEE Information Theory Society Paper Award, the Friedrich Wilhelm Bessel Research Award of the Alexander von Humboldt Foundation in 2007, the 2016 IEEE Eric E. Sumner Award "for pioneering contributions to the field of network coding," the 2018 ACM SIGMOBILE Test-of-Time Paper Award, the 2021 IEEE Richard W. Hamming Medal "for fundamental contributions to information theory and pioneering network coding and its applications," and the 2022 Claude E. Shannon Award. In 2015, he was named (together with Zhen Zhang) an Outstanding Overseas Chinese Information Theorist by China Information Theory Society. In 2018, with Shenghao Yang, he co-founded n-hop technologies in Hong Kong that has successfully deployed BATS code in Hong Kong Government's pilot smart lamppost system for wireless multi-hop transmission of sensor data. In 2019, his team won a Gold Medal with Congratulations of the Jury at the 47th International Exhibition of Inventions of Geneva for their invention "BATS: Enabling the Nervous System of Smart Cities." He has served on the committees for a number of information theory symposiums and workshops. He was the General Chair of the First and the Fourth Workshops on Network, Coding, and Applications (NetCod 2005 and 2008), the Technical Co-Chair of the 2006 IEEE International Symposium on Information Theory, the Technical Co-Chair of the 2006 IEEE Information Theory Workshop, Chengdu, China, and the General Co-Chair of the 2015 IEEE International Symposium on Information Theory. He serves as the Editor-at-Large for *Communications in Information and Systems*, an Editor of *Foundation and Trends in Communications and Information Theory* and *Foundation and Trends in Networking*, and was an Associate Editor for Shannon Theory of IEEE TRANSACTIONS ON INFORMATION THEORY from 2003 to 2005. From 2011 to 2012, he served as a Distinguished Lecturer for the IEEE Information Theory Society.